

# COHERENT AND INCOHERENT SPOUTING PROCESS IN SPOUTED BED: A COMBINED CFD AND EXPERIMENTAL STUDY

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**BACHELOR OF TECHNOLOGY**

By

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# Abstract

This research internship report presents a comprehensive investigation into the hydrodynamic behavior of a gas-solid system within a 2D rectangular spouted bed, using both **experimental** and **computational techniques**. The primary goal was to determine the **minimum spouting velocity  $U_{ms}$** , analyze the **spout dynamics**, and characterize the **transition between coherent and incoherent spouting regimes** under varying gas flow conditions.

The experimental setup comprised a transparent, pseudo-2D rectangular spouted bed with air supplied through a controlled nozzle, where pressure drop measurements and high-speed video recordings were employed to determine spouting behavior. Experimental results were validated through advanced **Computational Fluid Dynamics (CFD)** simulations performed in **ANSYS Fluent 2021 R2**, using the **Eulerian–Eulerian Two-Fluid Model (TFM)** and the **Gidaspow drag model**. The simulation used a transient solver with a standard  $k-\epsilon$  turbulence model and involved grid independence studies using 3 mm, 4 mm, and 5 mm mesh sizes. A **5 mm grid** was selected based on a trade-off between computational efficiency and solution accuracy.

Minimum spouting velocity was found to be **11.13 m/s experimentally** and **10 m/s numerically**, showing strong agreement between methods. **Pressure drop vs. gas velocity characteristics** were used to identify spouting transitions, and **volume fraction contours** from the CFD model validated the formation and structure of the spout. The study successfully distinguished **coherent spouting** (symmetrical and steady spout-fountain behavior) from **incoherent spouting** (irregular, fluctuating flow), supported by both simulation results and experimental observations.

Additionally, spatial profiles of **solid and gas vertical velocities, solid volume fractions**, and their variation with height and lateral position offered deeper insights into particle circulation mechanisms. The validated CFD model proves effective in predicting dynamic spouted bed behavior and offers a reliable tool for further industrial applications, design scale-up, and parametric analysis of spouted bed reactors.

Beyond the core analysis, this report also presents a comparative evaluation of CFD–DEM and TFM modeling approaches, a validation study of the Ergun equation for packed beds using TFM, and an investigation into jet meandering instability within a granular bed. These extended chapters (6–8) contribute additional perspectives on modeling fidelity, theoretical consistency, and complex flow behavior, supporting broader research and application in gas-solid systems.

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# 1 Introduction

## 1.1 Background

Spouted bed systems are a type of gas-solid contactor widely utilized in industries such as drying, coating, combustion, and granulation. Their advantage over conventional fluidized beds lies in their ability to handle larger, irregular, and denser particles with efficient gas-solid contact and reduced erosion [1][2]. The typical spouted bed design consists of a central gas inlet that creates a jet, spouting particles upwards through a core region, which then circulate downward along the annular zone [2].

One of the key parameters governing the operation of spouted beds is the minimum spouting velocity ( $U_{ms}$ ) — the lowest gas velocity at which a stable spout forms and is maintained [2]. It is defined as the minimum gas velocity required to maintain a stable spouting condition in a spouted bed [1]. The value of  $U_{ms}$  is obtained when the fountain just collapses while decreasing superficial gas velocity. Accurate prediction and measurement of  $U_{ms}$  are crucial for the stable design and scale-up of such reactors [2][4].

While several empirical and semi-empirical correlations for  $U_{ms}$  have been proposed in literature based on cylindrical and conical spouted beds [2], studies on 2D rectangular spouted beds remain relatively limited. Moreover, much of the existing work lacks rigorous validation through Computational Fluid Dynamics (CFD) and controlled experimental setups [4].

In this study, a 2D rectangular spouted bed configuration is investigated, with the aim of accurately determining the minimum spouting velocity through both simulation (using ANSYS Fluent) and experimental validation [4] and To characterize the difference between coherent (stable) and incoherent(unstable)

spouting regimes. The project combines numerical modeling and hands-on lab experimentation to better understand flow behavior, regime transition, and critical operating conditions [4].

To further broaden the scope of this research, additional chapters explore comparative modeling approaches (Chapter 6), validation of classical pressure-drop correlations using TFM (Chapter 7), and the dynamic flow instability known as meandering (Chapter 8). These extensions support a deeper understanding of spouted bed physics and CFD applicability in more complex granular flow phenomena.

## **1.2 Motivation**

Understanding and predicting the minimum spouting velocity ( $U_{ms}$ ) is essential for the safe and efficient operation of spouted bed systems. Operating below this velocity leads to erratic flow, particle stagnation, and reduced process efficiency. Conversely, excessive gas velocity results in higher energy consumption and particle attrition. Thus, accurate determination of  $U_{ms}$  is not only a scientific challenge but also a key industrial requirement.

Although cylindrical and conical spouted beds have been extensively studied, the 2D rectangular spouted bed, which offers better flow visualization and reduced wall effects, is underexplored in literature. Additionally, most studies focus on either simulations or experiments in isolation. Few have attempted to validate CFD-predicted  $U_{ms}$  values against experimental measurements in a controlled environment.

This research aims to address this gap by employing both CFD simulations and experimental methods to determine  $U_{ms}$  for a 2D spouted bed. It serves as a

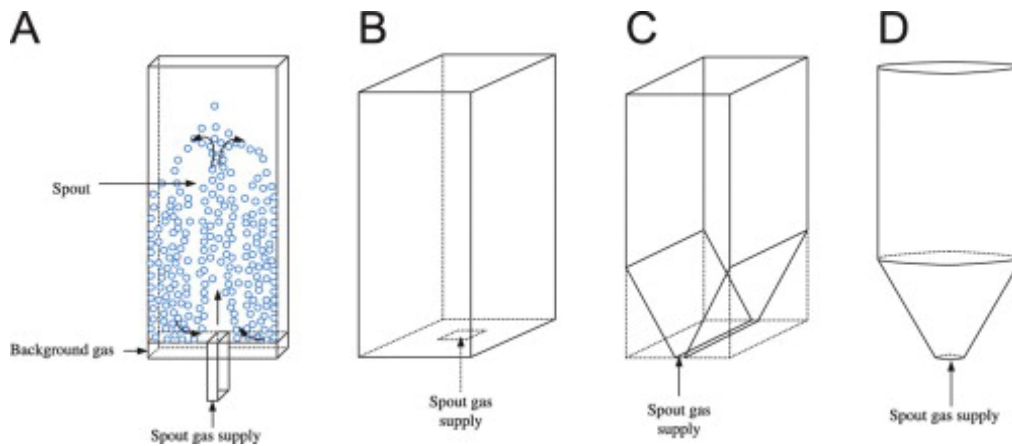
valuable opportunity to apply theoretical knowledge, CFD modeling skills, and experimental techniques, aligning with my long-term goal of pursuing a career in scientific research and development.

Building on this core objective, the project also explores comparative modeling (Chapter 6) to evaluate the trade-offs between TFM and CFD–DEM, applies classical theoretical validation techniques (Chapter 7) using the Ergun equation, and investigates advanced flow instabilities like meandering (Chapter 8). These components reflect a deeper motivation to develop holistic CFD competence and expand practical understanding of granular flow systems in complex regimes.

### **1.3 Spouted Bed**

Initially, spouted beds are designed with a single centered orifice to purge the fluid. Spouted beds are found to be efficient for coarse particles when compared with fluidised beds. Apart from this, coarse particles of irregular texture and sticky solids can be easily handled in spouted beds. The process of spouting leads to three major zones “Spout, Annulus and Fountain”. These zones are formed when the fluid is purged through the static bed with sufficient velocity. The spout is a fluid-particle region, aligned along the central axis of the bed, Annulus is the region filled with solid particles between the spout and bed wall and Fountain region is formed above the bed surface where the solid particles are lifted to a certain height.

There are different types of spouted bed configurations in Fig 1.1, such as pseudo two dimensional (Fig.1.1 A), rectangular with flat bottom (Fig.1.1 B) & taper bottom (Fig.1.1 C), and conical based cylindrical (Fig.1.1 D) which is also known as conventional or axisymmetric spouted bed. The following picture depicts the various types of SBs [5]



**Figure 1.1 Schematic representation of different types of spouted bed [5]**

## **1.4 Applications of Spouted Bed**

The spouted bed finds wide applications in industrial processes for an improved rate of particle circulation and thereby increased heat & mass transfer rates. Spouted bed provides an additional advantage of a reduction in bed pressure drop at operating conditions.

Because of these advantages, the spouted bed technology has been extensively used in various processes such as

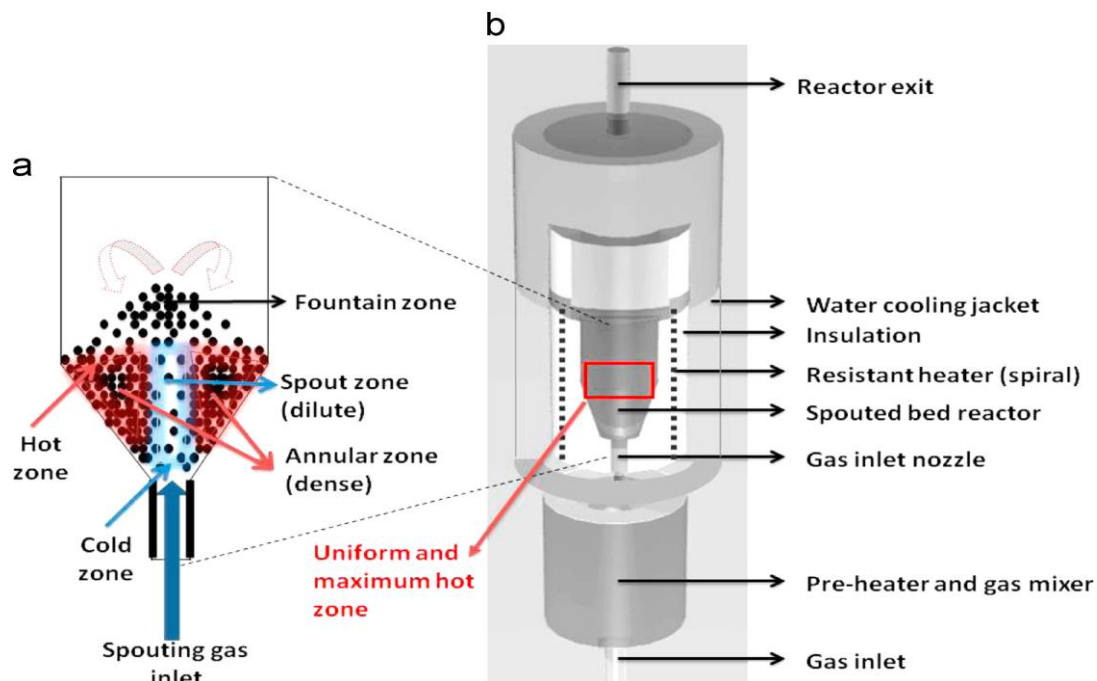
- Coating
- Drying of wet products
- Combustion
- Pyrolysis
- Gasification

### **1.4.1 Coating**

A lot of study has been done recently on using spouted beds for coating processes. The liquid or solution that needs to be coated is injected into the inlet gas and the particles are carried by spouted gas during the coating process. A



thin film of liquid is formed on the surface of the particles during this process, and the coating material sticks to the particles once the solvent evaporates. For the purpose of coating, a special type of spouted bed is used popularly known as Wurster coater [6]. Along with this, the spouted bed is found to be efficient in coating the nuclear fuel particles. Particles such as zirconia are coated with pyrolytic carbon can be performed in spouted bed via CVD. CVD of particles in spouted bed is depicted in the following Figure 1.2 [7]



**Figure 1.2: Schematic of (a) spouted bed reactor and (b) internals of coating unit [7]**

### 1.4.2 Drying of wet products

Spouted bed drying is a cost-effective and efficient method for drying paste-like materials such as foodstuffs, fertilizers, dyes, and chemicals.[8] The quality of the dried product is assessed by measuring moisture content, water activity , polyphenol degradation ratio , and mean powder diameter.[9]

The drying operation begins with the injection of heated air into the spouted bed. Upon reaching the desired temperature, the feed of concentrated extracts (CEs) is introduced at a preset flow rate along with atomizing air. [10]

Key operating variables, such as gas temperature, gas flow rate, and extract feed flow rate, significantly impact the drying rate, heat and mass transfer, and product properties. Optimal control of these variables ensures efficient drying and high quality product.

In this system, drying occurs through direct contact between the material film and hot air, with the dried film being mechanically removed from the inert surface by spouting. This method enhances drying efficiency and reduces product buildup, making spouted bed drying a valuable technique for various industrial applications.

### **1.4.3 Combustion**

The application of spouted bed reactors for combustion processes has not been extensively explored, despite their inherent advantages similar to fluidized beds, such as excellent gas-solids contact, ease of solids addition and withdrawal, and uniform bed temperature due to effective agitation. Arbib et al. (1981) investigated the use of spouted beds as combustors, focusing on the interplay between bed fluid mechanics and the combustion process.

Spouted bed reactors offer significant benefits over fluidized beds in various process applications, particularly in terms of contact efficiency, which is crucial for determining the extent of combustion. The intimate contact between gas and solids in spouted beds enhances combustion efficiency, making them a promising alternative for combustion processes that require uniform temperature distribution and effective gas-solids interaction.[11][12]

#### **1.4.4 Pyrolysis**

For over 20 years, laboratory-scale conical spouted bed reactors have been developed at the University of the Basque Country for pyrolyzing various solid wastes, including biomass, waste tires, and plastics. These reactors have demonstrated themselves as viable alternatives to conventional fluidized beds, offering superior solid mixing with low segregation and flexible gas flow distribution. Research by Olazar et al. [13] has shown that conical spouted beds can effectively process biomass particles, such as chips and shavings, without the need for inert materials. The reactor design is crucial for ensuring hydrodynamic stability and optimizing product formation during biomass pyrolysis, highlighting its importance in achieving efficient and effective pyrolysis processes.[6]

#### **1.4.5 Gasification**

Spouted beds are particularly relevant for biomass particle processing operations such as drying, toasting, pyrolysis, gasification, and combustion. In the context of gasification, spouted beds offer potential advantages due to their high gas-solid contact efficiency, superior heat and mass transfer rates, minimal spouting gas flow, high solid handling capacity, low particle entrainment, and short processing times. The gasification of solid fuels in spouted beds is a promising yet underutilized technique compared to other fluid bed configurations. The feasibility of biomass gasification using spouted bed reactors was examined through thermodynamic models, highlighting the innovative potential and applicability of spouted bed technology for efficient biomass gasification [6].

## 1.5 Problem Statement

Spouted bed systems are widely used for their excellent gas-solid contact efficiency, especially in processes involving coarse, irregular particles where conventional fluidized beds underperform. A key operating parameter in spouted beds is the minimum spouting velocity ( $U_{ms}$ ) which is the lowest superficial gas velocity required to initiate and maintain a stable spout of particles. Determining  $U_{ms}$  accurately is crucial for efficient operation, energy optimization, and scale-up of spouted bed reactors.

While several empirical and semi-empirical correlations exist for  $U_{ms}$  in conical or cylindrical spouted beds, they often do not generalize well to 2D rectangular spouted beds, which are commonly used in laboratory-scale studies for ease of visualization and measurement. Furthermore, most of these correlations fail to account for variations in particle properties, geometry, or inlet configurations. Additionally, prior numerical simulations using Eulerian-Eulerian Two-Fluid Models (TFM) have had limited experimental validation, especially for minimum spouting conditions.

Therefore, a combined CFD simulation (using TFM in ANSYS Fluent) and experimental approach is necessary to accurately predict and validate  $U_{ms}$  in a 2D spouted bed. This study seeks to address this gap by developing a validated numerical model and generating experimental data to compare and improve prediction accuracy.

In continuation, Chapter 6 addresses the challenge of choosing between accuracy and computational efficiency by comparing CFD–DEM and TFM. Chapter 7 identifies the need to validate CFD predictions with classical theory by applying the Ergun equation under packed bed conditions. Chapter 8 investigates the prediction and evolution of meandering flow instability, a

phenomenon rarely captured through simulations, but crucial for understanding jet behavior in granular media.

## 1.6 Objective

The primary objectives of this research internship project are:

- To investigate the hydrodynamic behavior of a 2D rectangular spouted bed under varying gas velocities.
- To determine the minimum spouting velocity ( $U_{ms}$ ) using both computational and experimental methods.
- To characterize the difference between coherent (stable) and incoherent (unstable) spouting regimes.
- To develop a comprehensive understanding through both experimental observations and CFD simulations
- To develop a Two-Fluid Model (TFM) simulation of the spouted bed in ANSYS Fluent, incorporating appropriate drag models, turbulence models, and multiphase flow settings.
- To design and conduct controlled experiments to measure  $U_{ms}$  for known particle and bed parameters.
- To validate the CFD-predicted  $U_{ms}$  values against experimental observations.
- Comparison between CFD DEM and TFM
- Validation of Ergun equation using TFM
- Analyze meandering flow instabilities using TFM

## **1.7 Outline of Report**

The structure of this research internship report is organized as follows:

### **Chapter 1: Introduction**

Provides the background, motivation, problem statement, and objectives of the study, along with the industrial relevance of spouted bed technology.

### **Chapter 2: Literature Review**

Summarizes existing work on spouted bed hydrodynamics, fluidization, spouting regimes, and previous computational and experimental methodologies.

### **Chapter 3: Spouted Bed Dynamics**

Discusses the various physical parameters such as spout diameter, minimum spouting velocity, and pressure drop correlations relevant to spouted bed operation.

### **Chapter 4: Methodology**

Describes the experimental setup and procedures used for data collection, as well as the CFD modeling approach, simulation settings, grid study, and post-processing techniques.

### **Chapter 5: Results and Discussion**

Presents and analyzes experimental and simulation results, including pressure drop curves, contour comparisons, velocity profiles, and spouting regime transitions.

Chapter 6: Comparison between CFD–DEM and TFM – Compares two different numerical modeling approaches for granular flows, evaluating their strengths, limitations, and applicability in spouted bed simulation.

Chapter 7: Validation of Ergun Equation Using TFM – Validates the Two-Fluid Model by comparing pressure drop results in a packed bed with predictions from the classical Ergun equation.

Chapter 8: Meandering – Analyzes jet instability in a granular bed using TFM, highlighting the development of meandering patterns and their relation to flow rate and bed structure.

Chapter 9: Conclusions and Future Work – Summarizes the key outcomes of the study and outlines future directions to extend the work in both computational and experimental domains.

## **2 Literature Review**

### **2.1 Spouted Bed**

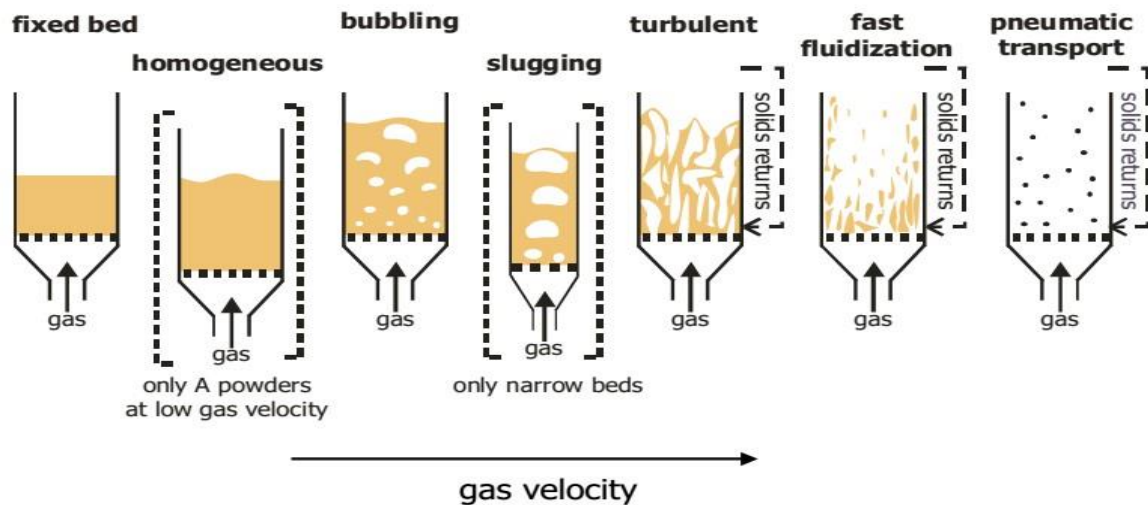
#### **2.1.1 Introduction to fluidization**

Fluidization is a process by which solid particles are transformed into a fluid-like state through suspension in a fluid, usually a gas or a liquid. This phenomenon occurs when the upward drag force from the fluid equals the downward gravitational force acting on the particles. At this point, known as minimum fluidization velocity, particles begin to behave collectively like a fluid. This transition marks the start of fluidization and is crucial for processes that require uniform mixing, efficient heat transfer, and controlled mass transport [1][14].

The majority of the fluidization applications have three important characteristics:-

- ❖ strong mixing within a fluidized bed results in the large interfaces between the fluid-solid phases
- ❖ excellent heat transfer between the fluidized bed and heat exchangers immersed in the bed
- ❖ fluidized particles can be easily transported between different reactors.





**Figure 2.1: Various forms of contacting of a batch of solids by fluid [14].**

### **Fluidization regimes:-**

As gas velocity increases beyond  $U_{mf}$ , the bed enters various flow regimes:

- **Fixed bed:** Particles are stationary; only interstitial gas flow occurs.
- **Particulate fluidization:** Smooth expansion with gentle particle movement.
- **Bubbling regime:** Gas escapes in the form of bubbles.
- **Slugging and turbulent fluidization:** Unstable, vigorous movement.
- **Pneumatic transport:** Particles are completely entrained and carried upward.

Each regime has specific industrial applications and is influenced by particle size, gas velocity, and bed geometry [1].

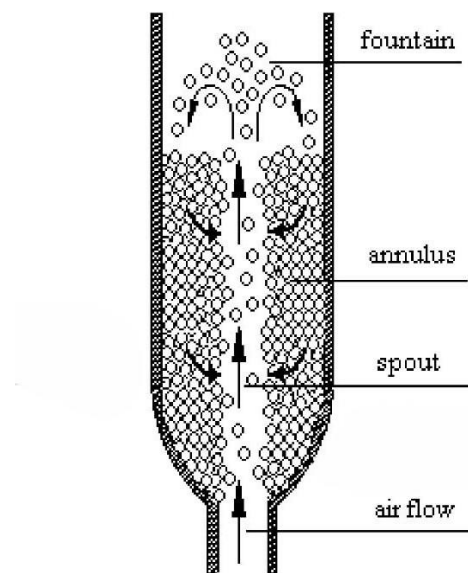
#### **2.1.2 Spouted bed**

A **spouted bed** differs from a conventional fluidized bed in that gas is introduced through a narrow central nozzle at high velocity. This forms a

vertical spout where particles are lifted and then fall back in the annular region, forming a stable recirculating loop . The system consists of three zones:

- **Spout:** Center jet, upward flow
- **Annulus:** Downward particle flow
- **Fountain:** Apex where particles reverse

Spouted beds are preferred when handling coarse, heavy, or irregular particles unsuitable for standard fluidization [2][3].



**Figure 2.2: Spouted bed [3]**

### 2.1.3 Fluidized bed vs Spouted bed [3]

| Feature          | Fluidized Bed               | Spouted Bed    |
|------------------|-----------------------------|----------------|
| Gas introduction | Uniform through distributor | Central nozzle |

|                    |                 |                             |
|--------------------|-----------------|-----------------------------|
| Particle motion    | Random bubbling | Cyclic, upward and downward |
| Suitable particles | Fine            | Coarse, dense               |
| Attrition          | Higher          | Lower                       |
| Mixing uniformity  | High            | Moderate                    |

**Table 2.1: Fluidized bed and Spouted bed comparison**

#### **2.1.4 Why Spouted bed?**

- Ease of handling Geldart D particles as the size of the particle is very high, it requires a very higher minimum fluidization velocity. To reduce this minimum fluidization velocity, we reduce the diameter at the bottom therefore there will a high velocity to fluidize the particles.
- To reduce operational pressure drop with improved particle recirculation rate and thereby increased heat and mass transfer rate
- Extensively used in various processes such as drying of wet product, combustion, pyrolysis, gasification etc [2][3].

## **2.2 Experimental Methodology**

The experiments were conducted in a pseudo 2D rectangular spouted bed made of transparent plexiglass, tapered at 70° up to a height of 14 cm and beyond 14 cm, the bed is 30 cm wide. The experimental setup is shown in Figure 3.1. The

particles used were glass beads with a diameter of 2.58 mm and a density of 2526 kg/m<sup>3</sup>. The experiments were carried out at various heights with rectangular inlet nozzle (30 mm × 20 mm) dimensions. Compressed air was supplied through the inlet nozzle, with flow rates measured using a rotameter. A dynamic pressure transducer (VARIOHM) was used to capture pressure fluctuations. This pressure transducer was calibrated with a Utube manometer. Data acquisition was performed at 1000 Hz via a National Instruments system, and the collected data were analyzed using LabVIEW software.



**Figure 2.3: Spouted bed**

### 3 Spouted Bed

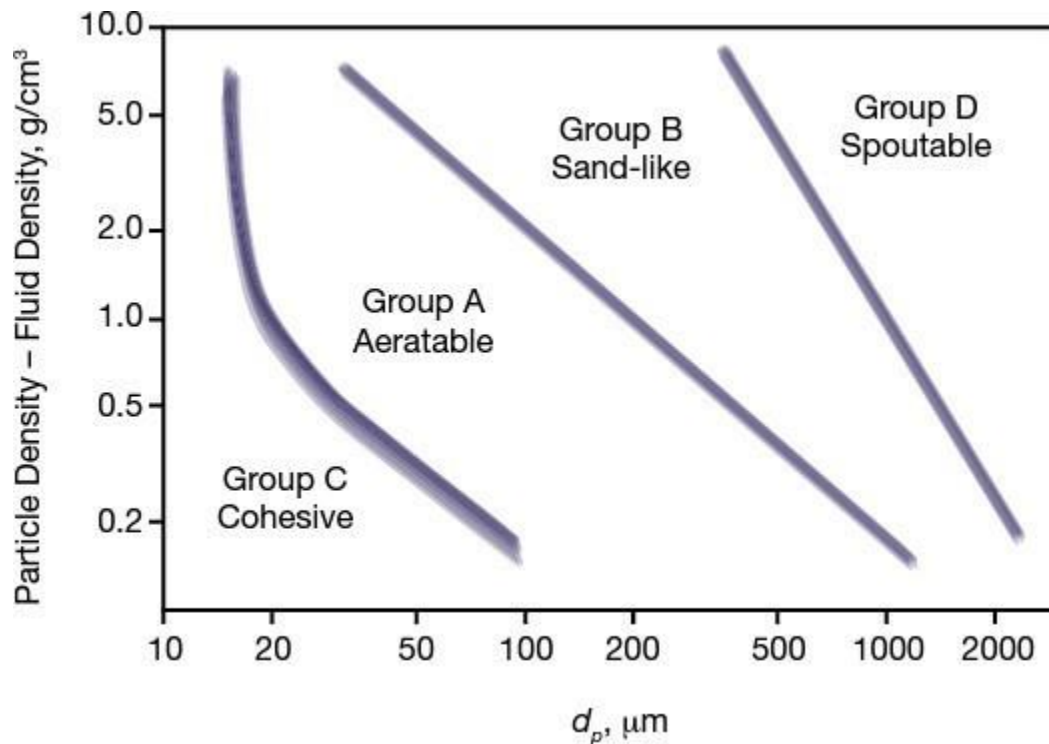
As mentioned before, the spouted bed is a specialized type of fluidized bed characterized by the introduction of gas through a single central orifice into a bed of granular material. This configuration creates three distinct zones within the bed: the spout, the fountain, and the annulus. The spout, located at the center of the bed, is a dilute phase where particles are fluidized by the upward flow of gas. Above the bed surface, the particles form a fountain as they are projected upwards before falling back down. The annulus, the region between the spout and the vessel wall, is where the particles circulate downward, completing the cycle. The main motivation was to reduce the operating pressure drop with improved particle recirculation rate [7][18].

Since the spouted bed technique was originally developed by Mathur and Gishler [4][15] for drying wheat, it has been successful in a range of operations including solid coating, pyrolysis and coal gasification. Different from conventional fluidized beds, the granular phase in spouted beds is agitated by the gases through a single nozzle, which causes the flow pattern of particles in spouted beds to be more regular than that in fluidized beds.

From the point of view of technological need, fundamental studies of flow pattern of granular solids are of great importance in designing spouted beds for optimal mixing. In addition, better understanding of the dynamics of spouted beds agitated by a single nozzle is crucial as a first step toward understanding the more complex.

Mathur and Gishler invented the spouted bed, while investigating a process to dry the coarse particles. Gledart [16], in his research has classified the particles into four categories based on the size and density of the particles. These include

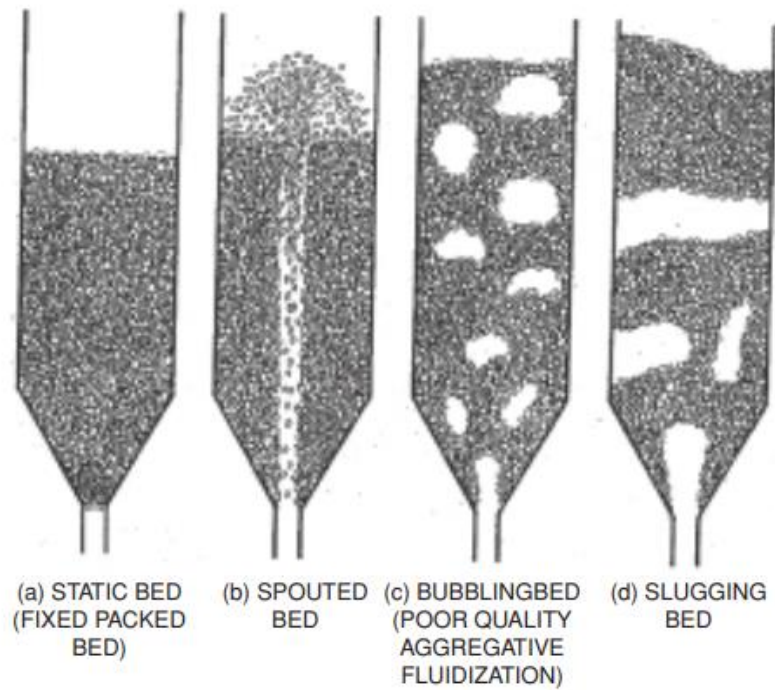
“Aeratable (Group A), Bubbly (Group B), Cohesive (Group C) and Spoutable (Group D)”. Hence, Coarse particles with the larger particle diameter and density falls under Group D classification, which are suitable for spouted beds. The following picture depicts the classification of spouted beds Figure 3.1.



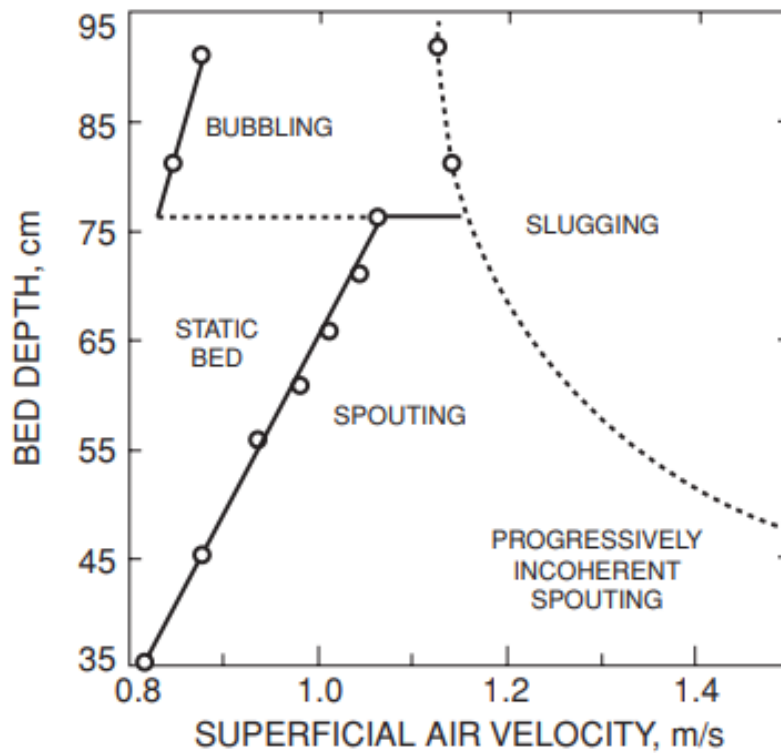
**Figure 3.1: Geldart Classification of Particles [17]**

### 3.1 Flow Regimes in Spouted Bed

Similar to fluidised beds, with the increase in inlet gas velocity regime transformation takes place in the spouted bed. As the superficial gas velocity increases, a transition from static bed to spouted bed and then to bubbling bed and slugging bed. In an early study on rectangular spouted bed, Dogan et al [19] characterised 8 different flow regimes. These include “Fixed bed, Internal jet spouting, Jet-in-Fluidized bed, Spouting, Dilute phase spouting, Transitional regime, Incoherent spouting and Slugging”. Flow transitions and flow regime map of the spouted bed are depicted below [24].



**Figure 3.2: Regimes Transitions in Spouted bed [2]**



**Figure 3.3: Flow regime map [2]**

## 3.2 Hydrodynamics of Spouted Bed

The hydrodynamics of spouted bed include the study of fluid and particle characteristics of spouted bed. These include “Minimum Spouting velocity, maximum spoutable depth, spout diameter, maximum pressure drops”. A few geometrical constraints are proposed for the stable spouting behaviour. The first constraint includes the ratio between fluid inlet diameter ( $D_i$ ) and the diameter of cylinder in case of conventional spouted bed ( $D$ ). For spouting to take place the fluid inlet velocity should be equal or greater than the terminal settling velocity of particles. From this we obtain a relation

$$\frac{\pi}{4} D^2 U_{mf} \leq \frac{\pi}{4} D_i^2 U_t \quad (3.2.1)$$

which further simplified as

$$\frac{D_i}{D} < \left( \frac{U_{mf}}{U_t} \right)^{1/2} \quad (3.2.2)$$

Here, according to Zakie-Richardson index in case of settling of particles

$$\frac{U_{mf}}{U_t} \simeq \epsilon_{mf}^n \quad (3.2.3)$$

Hence the equation 3.3.2 becomes

$$\frac{D_i}{D} \leq \epsilon_{mf}^{n/2} \quad (3.2.4)$$

Another constraint includes the ratio of fluid inlet diameter to the mean particle diameter. According to a study the ratio should not exceed 25-30, which is irrespective of particle density [25][26].



### 3.3 Minimum Spouting Velocity

It is defined as the minimum gas velocity required to maintain a stable spouting condition in a spouted bed, which can be determined visually when the fountain just collapses while reducing gas velocity [2] During the process of spouting  $U_{ms}$  is estimated when the pressure drop across the bed increases abruptly while decreasing the superficial gas inlet velocity. The minimum spouting velocity ( $U_{ms}$ ) is the minimum superficial inlet gas velocity at which there is an onset of spouting process in the bed [15]. If the gas velocity is below this critical velocity, there will be no motion in the bed. Above the minimum spouting velocity, drag force on the particle becomes sufficient enough to entrain the particles and establish a solid circulation. Minimum spouting velocity depends on the bed geometry like cone angle, column diameter, inlet diameter, particle properties like size and shape of the particles, polydispersity of the particle phase, etc. A large number of experiments have been performed to determine the minimum spouting velocity and a number of correlations have been proposed [15][27][28]. We have briefly presented a few of the correlations in table 3.1. It is worth mentioning that although a large number of correlations are

proposed their dependence on the parameters varies a lot and consequently, their prediction also varies significantly.

| Sl No | Correlations for minimum spouting velocity                                                                                                                                                                                                                                                                                               | Authors                   | Description                               |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------|
| 1     | $U_{ms} = \left(\frac{d_p}{D_c}\right) \left(\frac{D_i}{D_c}\right)^{\frac{1}{3}} \sqrt{\frac{2gH(\rho_s - \rho_f)}{\rho_f}}$                                                                                                                                                                                                            | Mathur and Gishler (1955) | $D_c < 0.6m$                              |
| 2     | $U_{ms} = k(1 - \varepsilon)^{1/2} \left(\frac{D_i}{D_c}\right)^2 \sqrt{\frac{2gH_0(\rho_s - \rho_f)}{\rho_f}}$                                                                                                                                                                                                                          | Madonna and Lama (1958)   | Theoretical                               |
| 3     | $U_{ms} = 2.44 \left(\frac{D_0}{D_c}\right)^{0.58} \left(\frac{H}{D_c}\right)^{0.5} \left(\frac{d_p}{D_c}\right)^{0.7} \left[\frac{2gH(\rho_s - \rho_f)}{\rho_f}\right]^{0.28}$                                                                                                                                                          | Anabtawi et al. (1992)    | For square column                         |
| 4     | $U_{ms} = \left(\frac{D_i}{D_c}\right)^{0.1} \left(\frac{d_p}{D_c}\right) \left[\frac{2gH(\rho_s - \rho_f)}{\rho_f}\right]^{0.28}$                                                                                                                                                                                                       | Olazar et al. (1994)      |                                           |
| 5     | $U_{ms} = 3\sqrt{2gH} \left(\frac{\alpha}{\lambda}\right) \left(\frac{\alpha}{d_p\phi}\right)^{-0.75} \left(\frac{H}{\alpha}\right)^{0.16} \left[\frac{\rho_s - \rho_f}{\rho_f}\right]^{0.23}$                                                                                                                                           | Costa and Taranto (2003)  | For square inlet                          |
| 6     | $U_{ms} = 0.38 \left(\frac{d_p^{3.48}}{D_c}\right)^{0.23} \left(\frac{H}{D_c}\right)^{0.3} \left[\frac{\rho_s - \rho_f}{\rho_f}\right]^{0.67}$                                                                                                                                                                                           | Zhong et al. (2013)       | $\frac{H}{D_c} > 1.5$ & $\phi = 60^\circ$ |
| 7     | $\frac{U_{ms}}{\sqrt{gH}} = 0.0526 \left[\left(\frac{\alpha}{\beta}\right)^{0.122} \left(\frac{\alpha}{\lambda}\right)^{-0.727} \left(\frac{\lambda}{d_p}\right)^{-0.689} \left(\frac{H}{d_p}\right)^{-0.081} \left(\frac{\rho_s - \rho_f}{\rho_f}\right)^{0.5}\right] \left[10.614 - \tan\left(\frac{\gamma}{2}\right)^{-0.934}\right]$ | Golshan et al. (2017)     | Including effect of cone angle            |

**Table 3.1: Correlations for minimum spouting velocity (U<sub>ms</sub>)**

### 3.4 Spout Diameter

One of the important parameter for spouted bed is the average spout diameter which plays an important role in determining the flow distribution between the spout and the annulus. McNab,G,(1972) [29] proposed a correlation which

states that the spout diameter ( $D_s$ ) varies with the superficial gas velocity and the inlet diameter by the following relation

$$D_s \propto U^{0.5} \cdot D_c^{0.68} \quad (3.4.1)$$

Green and Bridgwater [30] observed that the above correlation underestimates the spout diameter by 15 – 30%. However, the prediction shows a maximum deviation of 10% with the measured data of He et al. (1992) [31] when the column diameter is 91cm. Using force balance at the spout-annulus interface, Bridgwater and Mathur (1972) [32] found that the spout diameter,

$$D_s \propto G^{0.5} \cdot D_c^{0.75} \quad (3.4.2)$$

The authors considered both the hydrodynamics forces and the solid stress in their force balance and assumed that there is an insignificant variation of the spout diameter along the bed height. Later many scientists came up with a modified and new correlation for the spout diameter. Some of the correlations proposed for the spout diameter are mentioned in the following

| Author                   | Correlation Developed                                                                        | Types of Spouted Beds |
|--------------------------|----------------------------------------------------------------------------------------------|-----------------------|
| Malek et al [20]         | $D_s = (0.0793 \log D_a + 0.105) G^{0.5a}$                                                   | Cylindrical           |
| Lefroy and Davidson[45]  | $D_s = \frac{1.06(D_c)^{2/3}}{d_p^{1/3}}$                                                    | Cylindrical           |
| Bridgwater & Mathur [32] | $D_s = \frac{0.2866 G^{0.5} \cdot D_c^{0.75}}{\rho_b^{0.26a}}$                               | Cylindrical           |
| Wu et al [46]            | $D_s = \frac{5.61 G^{0.433} \cdot D_c^{0.583} \mu_g}{(\rho_b \cdot \rho_g \cdot g)^{0.283}}$ | Cylindrical           |

|                 |                                                                                                                             |             |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------|-------------|
| Hook et al [47] | $\frac{D_s}{D_c} = \left( \frac{2.028 \rho_g \cdot \mu_i^2}{(\rho_p - \rho_g)(1 - \varepsilon_m)g \cdot D_c} \right)^{1/4}$ | Cylindrical |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------|-------------|

**Table 3.2 Correlations developed for Spout diameter**

### 3.5 Maximum Spoutable Bed Height

The maximum spoutable particle bed height,  $H_m$  is the maximum initial static bed height at which regular spouting pattern can be observed.  $H_m$  is one of the most important scaling parameter used in the design and scale-up of a spouted bed. It is defined as “the maximum height of the initial static bed, beyond which stable spouting is not observed”. If the initial height of the bed is greater than the maximum spoutable bed height, with the purging of gas the bed undergoes slugging and bubbling. Hence, for the successful operation of a spouted bed, it is required to obtain a better correlation for the maximum efficiency of the spouted bed. Many correlations were proposed for estimating the maximum spoutable bed height which depends on the particle and fluid characteristics together with the bed shape. The following table 3.3 contained the correlations available in the literature

| <u>Sr.No.</u> | Author                   | Correlation for Maximum <u>Spoutable</u> Bed Height                                                                                                                  |
|---------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01            | Malek & Lu [20]          | $\left(\frac{H_m}{D_c}\right) = 418 \left(\frac{D_c}{d_p}\right)^{0.75} \left(\frac{D_c}{D_i}\right) \rho_p^{-1.2} \Phi^{-2a}$                                       |
| 02            | Littman et al<br>[21]    | $\frac{H_m \cdot D_s}{(D_c^2 - D_s^2)} = 0.345 \left(\frac{d_s}{D_c}\right)^{-0.384}$                                                                                |
| 03            | Littman et al<br>[22]    | $\frac{H_m \cdot D_i}{D_c^2} = 0.218 + \frac{0.005}{A} \quad A > 0.02$                                                                                               |
| 04            | Morgan III et al<br>[23] | $\frac{H_m \cdot D_i}{D_c^2} = 0.218 + \frac{5.13 \times 10^{-3}}{A} + \frac{2.54 \times 10^{-5}}{A^2} \quad A > 0.014$<br>$= 175(A - 0.01) \quad 0.010 < A < 0.014$ |

**Table 3.3 Correlations developed for maximum spoutable bed height**

### **3.6 Peak pressure drop & Spouting pressure drop in Spouted Bed**

The important parameters associated to the design aspects of the spouted bed are peak ( $\Delta P_{\text{peak}}$ ) and spouting ( $\Delta P_s$ ) pressure drops. The peak pressure drop is defined as the maximum pressure drop across a granular bed during operation. Once a regular spouting is established, the operational pressure drop is termed as the spouting pressure drop.  $\Delta P_{\text{peak}}$  is inevitably greater than the operational spouting pressure drop [33] This is due to the fact of additional force required to break the inter-particle force due to interlocking before the formation of a cavity near the gas entry [3]. A large number of studies are reported with a focus on the effect of bed geometry on operating conditions. They have developed empirical correlations to find the maximum pressure drop ( $\Delta P_{\text{peak}}$ ) for conical [34,35], cylindrical [32,36] and flat bottom cylindrical

spouted bed [37,38]. A very few studies are available on the rectangular spouted bed. Rocha (1995) [39] used a two-dimensional rectangular spouted bed in their study and proposed a correlation to calculate the maximum pressure drop. Yang, (2020b) [40] have also proposed a similar empirical correlation to calculate the maximum pressure drop in a rectangular spouted bed for Geldart B & D particles.

## 4 Experimental Methodology and Computational Model

### 4.1 Experimental Methodology

The experiments have been performed in a 2D rectangular spouted bed, made of transparent plexiglass. The schematic outline of the experimental setup is shown in Figure 4.1. The particles used were glass beads with a diameter of 2.58 mm and a density of 2526 kg/m<sup>3</sup>. A packed bed of constant height (Here H<sub>s</sub>=14cm for all cases unless it is specified) has been prepared by pouring the particles under gravity. Compressed air is supplied from an inlet nozzle (dia 1 inch) to the air chamber. This air then enters the bed through mesh arranged in between flanges located at the lowest position of the tapered section. The dimensions of the air entry region is 30mm x 20mm. Minimum spouting velocity is defined based on the air entry cross-sectional area. A dynamic pressure transducer (VARIOHM) was used to capture pressure fluctuations. This pressure transducer was calibrated with a Utube manometer. Data acquisition was performed at 1000 Hz via a National Instruments system, and the collected data were analyzed using LabVIEW software.

#### 4.1.1 Minimum Spouting Velocity

The pressure drop characteristic curve was used to predict the minimum spouting velocity. The average of 1000 time series data of pressure drop taken at 1 second was used [4]. For N data points of the pressure drop  $\Delta P_i$  collected by LabVIEW, where  $i = 1, 2, \dots, N$ , the average pressure drop  $\Delta \bar{P}$  was calculated as:

$$\Delta \bar{P} = \frac{1}{N} \sum_{i=1}^N \Delta P_i$$

Here,  $\Delta P_i$  represents each individual pressure drop across the bed.

The average pressure drop was measured like this for varying superficial gas velocity somewhat beyond the point where stable spouting starts. Then again superficial gas velocity was decreased and average pressure drop was evaluated. This data of varying superficial gas velocity and pressure drop corresponding to it was used to predict the minimum spouting velocity. Along with this it was also taken care that minimum spouting velocity value comes approximately equal to the value of superficial gas velocity where fountain collapse [4].

#### **4.1.2 Particle image velocimetry (PIV) for Particle phase velocity measurement**

A pseudo two dimensional spouted bed has an advantage to take picture from the frontal face and to analyze them to predicted the particle velocity. The particle image velocimetry (PIV) technique was used to predict the velocity of particles. Phantom (MIRO LAB140), a high speed camera along with LED lights (VERITAS, model:1417- 700, 120W intelligent LED luminaire) was used to capture frames at velocity equal to 1.2 to 2 times the minimum spouting velocity. A thousand frames were captured by the camera at a rate of 419 pictures per second (pps) with a resolution of  $2048 \times 1536$  and an exposure time of 800  $\mu$ s. There framed were analyzed in PIV package in MATLAB. Steps involved in PIV analysis are as follows :

1. Load the captured images in Time Resolve Sequencing (i.e., [A + B], [B + C], . . .) and define region of interest (ROI) from image settings. Make sure to define region where particle analysis is needed to avoid white background which creates unnecessary noise during analysis.
2. Define masks to exclude region from analysis from image settings.
3. Perform the analysis by Fast Fourier Transform window deformation algorithm with 128, 64 and 32 interrogation area.



4. Perform calibration to convert pixels/frame into m/s.
5. Using post-processing, remove the false/dummy vector generated due to white background.
6. Find the mean frame and we are ready to extract the particle phase dynamics.



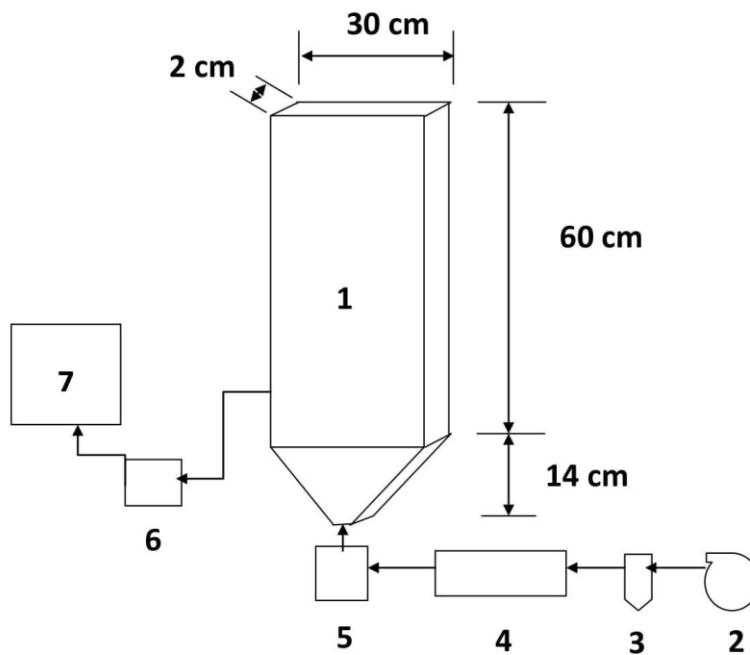
**Figure 4.1: Experimental setup.**

## **4.2 Computational Methodology**

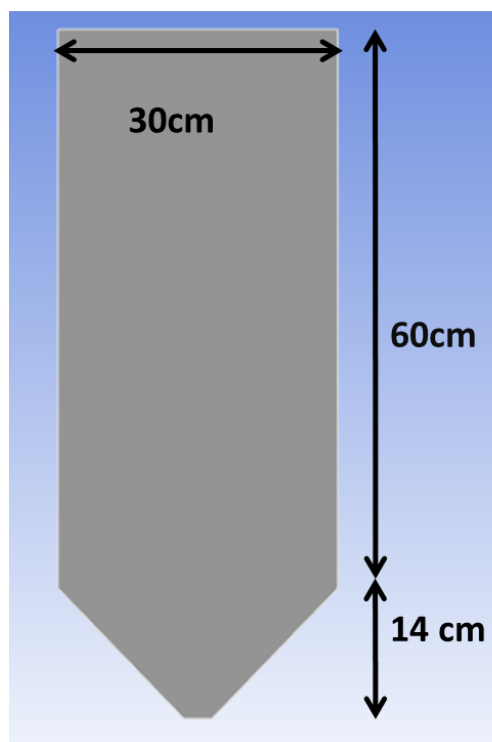
This section details the computational fluid dynamics (CFD) approach used to simulate the gas-solid flow in a 2D rectangular spouted bed using ANSYS Fluent 2021 R2, based on the Eulerian-Eulerian Two-Fluid Model (TFM) [41]. The objective was to numerically predict the minimum spouting velocity ( $U_{ms}$ ) and associated hydrodynamic parameters, and validate them experimentally.

### **4.2.1 System Geometry**

The simulations are performed in a tapered rectangular spouted bed. The domain used for the present study is 0.3m (width) x 0.74m (height) with a thickness of 0.02m. The schematic of the geometry of rectangular spout bed is shown in Figure 4.2 and the front view of numerical geometry with mesh is shown in Figure 4.3. The initial bed height was 0.14m ( $H_s$ ). The number of particles used in the present simulations corresponds to the physical dimension of the system used for experiments and also of the order or larger than that used in earlier studies [42]. The geometry was constructed using ANSYS DesignModeler, and three different unstructured quad meshes were created in ANSYS Meshing to perform a grid independence study. The physical properties of the particle is shown in Table 4.1. The parameters used in the current simulation are given in Table 4.2.



**Figure 4.2: Schematic of rectangular spouted bed (1:Spout bed, 2:Air compressor, 3:Moisture separator and Air filter, 4:Rotameter, 5:Air chamber, 6:DAQ system, 7:Monitor)**



**Figure 4.3: Front view of simulation bed geometry**

| Material            | Sieve size (mm) | Particle density (kg/m <sup>3</sup> ) | Geldart classification | Angle of repose |
|---------------------|-----------------|---------------------------------------|------------------------|-----------------|
| Glass beads (GB258) | -2.80+2.36      | 2526                                  | D                      | 28.6 ± 1.3      |

**Table 4.1: Physical properties of particle used**

| Parameters                                                        | Value                    |
|-------------------------------------------------------------------|--------------------------|
| Bed dimensions                                                    | 0.3m x 0.02m x 0.74m     |
| Initial static bed height (H <sub>s</sub> , m)                    | 0.14                     |
| Minimum spouting velocity (U <sub>ms</sub> , m/s)                 | 10 (Simu) & 11.13 (Expt) |
| Inlet superficial gas velocity (U <sub>g</sub> /U <sub>ms</sub> ) | 1.2                      |
| Particle density ( $\rho_p$ , kg/m <sup>3</sup> )                 | 2526                     |
| Particle diameter ( $d_p$ , mm)                                   | 2.58                     |
| Particle restitution coefficient                                  | 0.97                     |
| Particle Young's modulus (Pa)                                     | 1 x 10 <sup>9</sup>      |
| Ansys timestep (s)                                                | 1 x 10 <sup>-4</sup>     |

**Table 4.2: Simulation parameters used in the current study**

### 4.2.2 Grid Independence Study

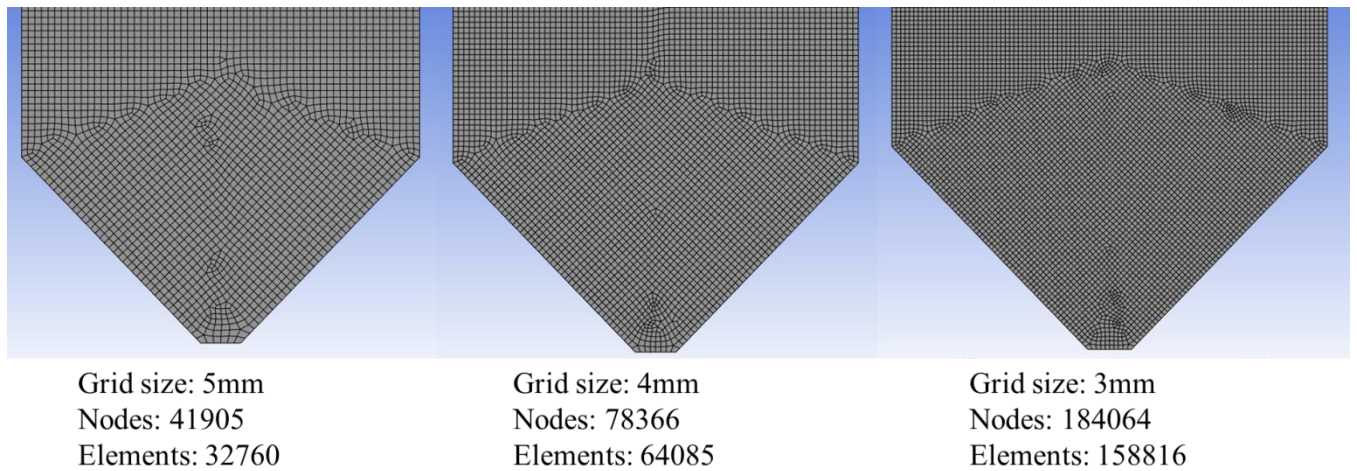
Even with the advent of high speed computational facilities, a fully (particle) resolved ANSYS simulation with few thousands of particles for an experimentally realizable system size is still beyond the reach. Therefore, the research in the area of spouted and fluidised bed include the interphase exchange of the momentum through the calculation of the forces at the particle centre.

To obtain an optimum grid for the present simulations a grid independence study with three sets of computational grids are performed 5mm,4mm,3mm. The aim was to determine whether finer meshes significantly affect simulation results or whether coarser meshes can provide similar accuracy with lower computational cost.

The structured mesh was generated in ANSYS Meshing, with physics preference set to CFD and solver preference to Fluent. The following mesh statistics were used:

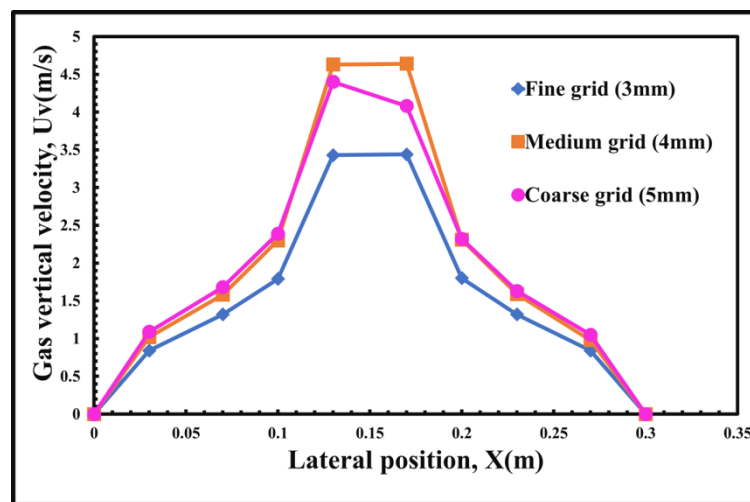
| Mesh type | Element size | Nodes   | Elements |
|-----------|--------------|---------|----------|
| Fine      | 3mm          | 184,064 | 158816   |
| Medium    | 4mm          | 78,366  | 64085    |
| Coarse    | 5mm          | 41,905  | 32760    |

**Table 4.3: Different grid size table**

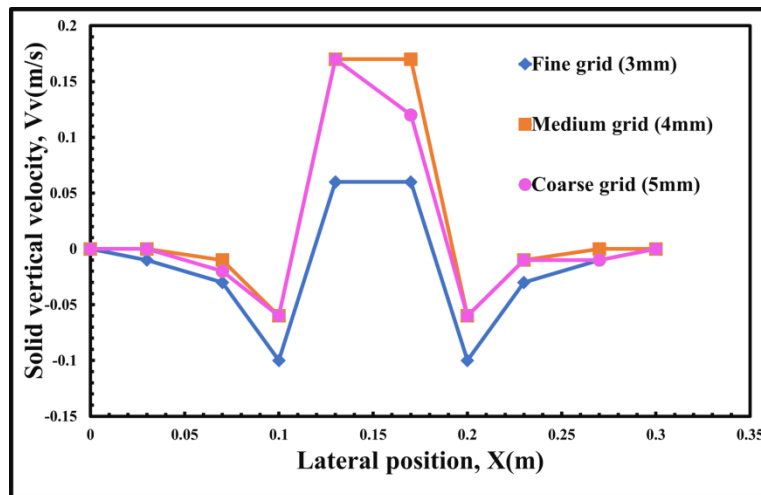


**Figure 4.4: Grid size comparison**

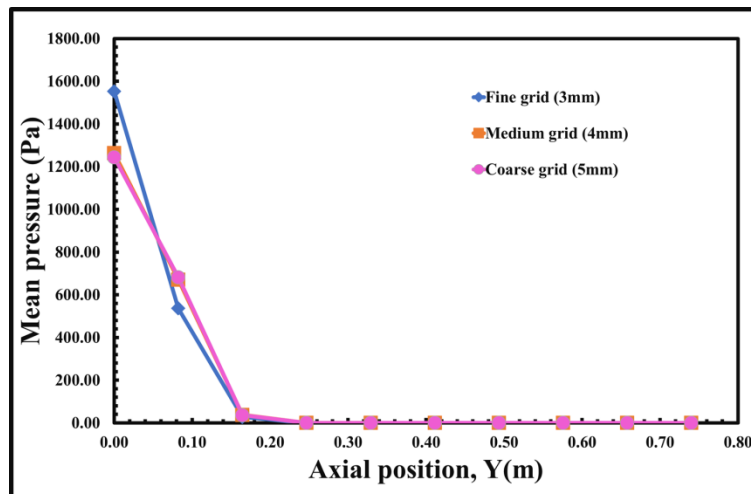
A common method to assess grid sensitivity is by comparing a representative flow parameter. In this case, gas and solid vertical velocity across the lateral position and mean pressure across axial position at a height ( $H_s=0.14\text{m}$ ) in the spouted bed was plotted for all three grids as shown in figure 4.4 (a), (b), (c).



**(a)**



(b)



(c)

**Figure 4.5: Lateral distribution of (a) gas vertical velocity (b) solid vertical velocity (c) Axial distribution mean pressure drop; Bed height  $Y=1.0H_s$ , Superficial gas velocity  $U_g = 14\text{m/s}$ .**

### Observation and Interpretation:

As seen in the graphs:

- The 3 mm grid (fine) shows a slightly lower peak  $U_v$  and  $V_v$  and more spread in velocity distribution, likely due to higher resolution and smoother gradients.

- The 4 mm and 5 mm grids show very similar peak velocities and mean pressure.
- The deviation between 4 mm and 5 mm is negligible in critical flow features like spout height, vertical velocity and mean pressure, indicating convergence of solution.

This behavior is consistent with literature where a coarser grid may still be acceptable if it captures the key flow dynamics within engineering tolerance [1][2]. According to Fluent best practices, once mesh refinement leads to less than ~5% variation in output, the simulation is considered mesh-independent [1].

#### **Choice of Grid for Final Simulation:**

Although the 3 mm mesh provided slightly finer flow details, it also resulted in significantly higher computational time and data storage requirements. On the other hand, the 5 mm grid showed results comparable to the 4 mm mesh, particularly in peak velocity, mean pressure and spout profile, and thus was selected for the final simulations.

This choice balances accuracy and computational efficiency, as using a coarser grid, reduces clock time per time step and Enables easier postprocessing and data handling

Therefore, the 5 mm mesh was adopted for all further analyses to ensure timely simulation turnaround without compromising predictive quality.

### **4.2.3 Solver and Multiphase model setup**

The simulations used the Eulerian multiphase model with the Two-Fluid Model (TFM) in transient mode in ANSYS with velocity inlet and pressure outlet conditions [41]. Transient setting table is shown is table 4.3.



The governing equations solved for gas (q) and solid (s) phase included continuity and momentum,

### **Governing Equations:**

**Continuity equation:** It is same for both the phases

$$\frac{\partial(\alpha_q \rho_q v_q)}{\partial t} + \nabla \cdot (\alpha_q \rho_q v_q) = 0$$

### **Momentum equations:**

Fluid phase:

$$\frac{\partial(\alpha_q \rho_q v_q)}{\partial t} + \nabla \cdot (\alpha_q \rho_q v_q) = -\rho_q \nabla p + \nabla \cdot \overline{\tau}_q + \alpha_q \rho_q g + K_{sq}(v_q - v_s) + F$$

Solid phase:

$$\frac{\partial(\alpha_s \rho_s v_s)}{\partial t} + \nabla \cdot (\alpha_s \rho_s v_s) = -\rho_s \nabla p + \nabla p_s + \nabla \cdot \overline{\tau}_s + \alpha_s \rho_s g + K_{sq}(v_q - v_s) + F$$

Here the variables are defined as  $\alpha$ ,  $\rho$ ,  $v$ ,  $\nabla p$ ,  $\tau$ ,  $K_{sq}$ ,  $\nabla p_s$ ,  $F$ , volume fraction of the particular phase, density, velocity, pressure drop, stress tensor, interphase momentum exchange coefficient, solid pressure drop and forces due to lift, virtual mass etc. for the two phases Gas (q) and solid (s).

### **Gidaspow Drag Model (1994)**

The Gidaspow drag model is one of the most widely used models for computing the interphase momentum exchange (drag force) between gas and solid phases in multiphase flow simulations, particularly within the Eulerian–Eulerian Two-Fluid Model (TFM) framework [41].

Proposed by D. Gidaspow (1994) [43], this model is a hybrid drag law that blends two well-known empirical correlations:

- ❖ Wen–Yu correlation: for dilute systems (solid volume fraction  $< 0.2$ )
- ❖ Ergun equation: for dense systems (solid volume fraction  $\geq 0.2$ )

This blending ensures accurate drag force estimation across both dilute and dense regimes of particle concentration, making it suitable for systems like spouted beds, fluidized beds, and pneumatic transport [14][43].

- The interphase momentum transfer coefficient expressed as:

$$K_{sq} = \begin{cases} \frac{3}{4} \frac{c_D \alpha_q \alpha_s \rho_q |v_q - v_s|}{d_p} & \alpha < 0.2 \\ 150 \frac{\mu_g \alpha_s^2}{\alpha_q^2 d_p^2} + 1.75 \frac{\rho_q \alpha_s}{\alpha_q d_p} |v_q - v_s| & \alpha \geq 0.2 \end{cases}$$

- The drag coefficient  $c_D$  is calculated by:

$$c_D = \begin{cases} \frac{24}{Re} [1 + 0.15(Re)^{0.687}] & Re < 1000 \\ 0.44 & Re \geq 1000 \end{cases}$$

- The particle Reynolds number  $Re$  defined as:

$$Re = \frac{\rho_q d_p |v_q - v_s|}{\mu_q}$$

Here,  $\mu$  is the dynamic viscosity and  $d_p$  is the diameter of particle.

## k-ε Turbulence Model

This model is used for capturing the gas phase turbulence. It solves for the turbulent kinetic energy (k) and turbulence dissipation rate (ε) [44].

Transport equations for k-ε Model:

1. Turbulence kinetic energy (k) equation:

$$\frac{\partial(\rho k)}{\partial t} + \nabla \cdot (\rho k u) = \nabla \cdot \left( \nabla k \frac{\mu_t}{\sigma_k} \right) + G_k - \rho \varepsilon$$

2. Dissipation rate (ε) equation :

$$\frac{\partial(\rho \varepsilon)}{\partial t} + \nabla \cdot (\rho \varepsilon u) = \nabla \cdot \left( \nabla \varepsilon \frac{\mu_t}{\varepsilon} \right) + C1 \frac{\varepsilon}{k} G_k - C2 \rho \frac{\varepsilon^2}{k}$$

| Parameter                      | Value                 |
|--------------------------------|-----------------------|
| Time step size                 | 0.0001                |
| Total time steps               | 30000                 |
| Maximum iteration per timestep | 20                    |
| Residual criteria              | $10^{-5}$             |
| Auto-save                      | Every 1000 time steps |
| Profile/Reporting interval     | 1                     |

**Table 4.4: Table of transient setting**

#### 4.2.4 Initialization and Patching

**Standard Initialization** in ANSYS Fluent is a built-in method used to assign initial values to all solution variables before starting the simulation. It is especially important in transient simulations, where the initial conditions can influence how quickly and accurately the solution converges.

In this project, Standard Initialization was performed from the velocity inlet boundary, ensuring that:

- Gas velocity was initialized along the vertical axis.
- Pressure was adjusted to hydrostatic conditions.
- Volume fractions were uniformly initialized, except for the packed bed region (which was patched separately).

This approach offers a fast and stable starting point for the solver, improving convergence rate and reducing residual oscillations during the early time steps. While more advanced initialization options like Hybrid Initialization or Manual Initialization are available, Standard Initialization is often sufficient for Eulerian multiphase simulations where geometry and flow directions are well-defined [41].

After initialization, the solid volume fraction was patched to **0.6** in the entire bed up to **14 cm height**, representing the packed bed up to the conical section. This was done using the Patch tool in Fluent, based on selected cell zones.

#### 4.2.5 Post-Processing

During Simulation:

Contour plots of solid volume fraction were saved as animations every 50 time steps.

#### After Simulation:

- The last .dat and .cas files were opened in CFD-Post.
- Contour slices were made using vertical and horizontal planes to visualize volume fraction, velocity, and pressure.
- Line probes were placed along the bed height to extract data for plotting spout height and pressure profiles.

## 5 Results & Discussion

In the present work, the hydrodynamic behavior of a gas-solid system for monodispersed spherical glass beads (GB258) has been presented systematically. As a first step of the study, the minimum spouting gas velocity ( $U_{ms}$ ) is predicted numerically using the pressure drop characteristic curve and validated with our experimental findings. Then, the effect of increasing superficial gas velocity ( $U_g$ ) on particle dynamics has been studied.

### 5.1 Prediction of Minimum Spouting Velocity

Experiments and numerical simulations have been carried out simultaneously to predict the minimum spouting velocity. The pressure drop ( $\Delta P$ ) characteristic curves for monodispersed glass beads are shown in Figure 5.1. Experiments have been conducted for a particular particle-bed height ( $H_s=14$  cm) as shown in figure 5.2. It is observed that mean  $\Delta P$  across the bed increases with an increase in superficial gas velocity ( $U_g$ ), as expected for a packed bed system. Then it reaches to a maximum value  $\Delta P_{max}$  for a certain gas velocity before the onset of spouting. Increasing  $U_g$  beyond this point results in a sharp decrease in the bed pressure drop. The supplied airflow at the gas inlet is sufficient enough to lift the particles at the gas entry point, which forms a cavity. At this flow rate there is also an increase in effective bed voidage. With further increase in  $U_g$ , pressure drop starts decreasing due to a further increase in bed voidage. A simultaneous expansion in the cavity and formation of a jet in the flow direction is also observed. With further increase in gas velocity, the phenomena of jet collapsing and bubble formation occur, which lead to the formation of a dilute zone in the central part of the bed. The particles move upward, sometime in the form of cluster, in the dilute zone. Continuous oscillation of jets and formation

of bubbles are observed. Bubbles reach to the surface of the bed, burst there and disperse the particles back to the surface of the bed. Beyond this point, the pressure drop across the bed becomes independent of gas velocity. In the next step, the bed behavior has been investigated with decreasing gas velocity. It is observed that there is a sudden (quite insignificant) jump in pressure drop due to a sudden collapse of the fountain. The gas flow rate corresponds to this point is defined as the minimum spouting velocity  $U_{ms}$ . With a further decrease in  $U_g$ , the pressure drop starts decreasing, and finally, the bed behaves like a packed bed showing a marginal bed expansion. The pressure drop during ascending gas flow rate is always higher than that for descending gas flow rate. This is because the particles are loosely packed in the latter case. In the present investigation,  $U_{ms}$  was found to be 11.13m/s from experimental results and 10m/s from the numerical simulations. It is worth noting that the prediction of minimum spouting velocity from the pressure drop with decreasing gas velocity needs higher resolution as this jump in the pressure drop is much lower than the maximum pressure drop that occurs during the experiment with ascending gas velocity. In the case of numerical simulation, confirmation of the predicted  $U_{ms}$  can be made through the contour plots of the void fraction in the bed as shown in Fig. 5.3. Variation of minimum spouting velocity with bed height has been predicted numerically and compared with the experimental prediction as shown in Fig 5.1.

Trends for the average bed pressure drops for both the experimental and simulated one are found identical however, a quantitative difference might have originated from the initial packing configuration and due to the assumptions in ANSYS Fluent's Two Fluid Model. Nevertheless the numerically obtained spouting pattern is similar to experimental results and numerically predicted Minimum Spouting velocity ( $U_{ms}$ ) also compares well with the experimental prediction.

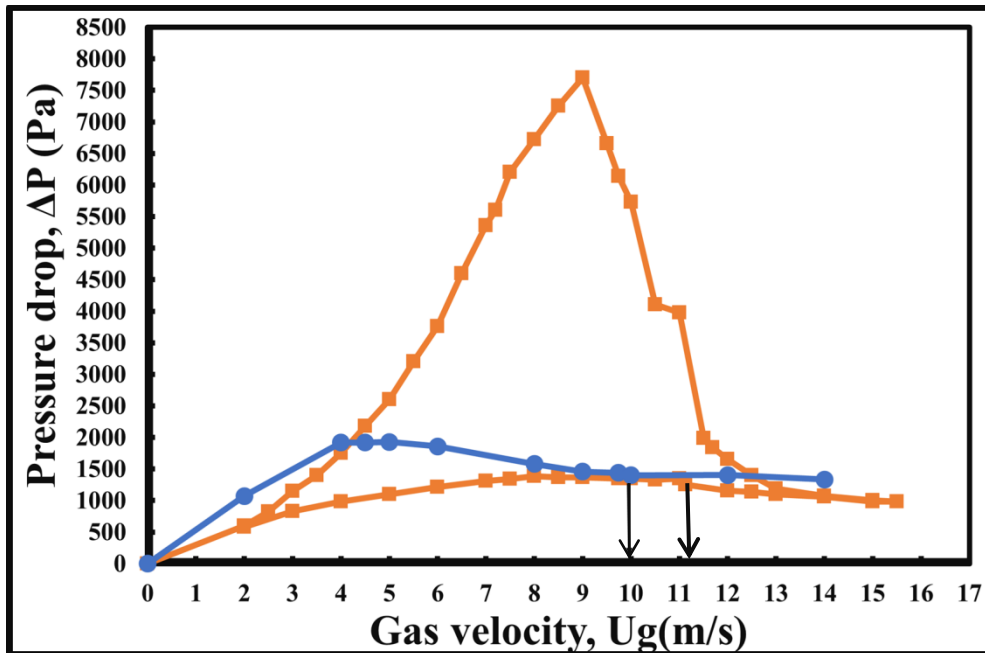


Figure 5.1: Pressure drop characteristics curve

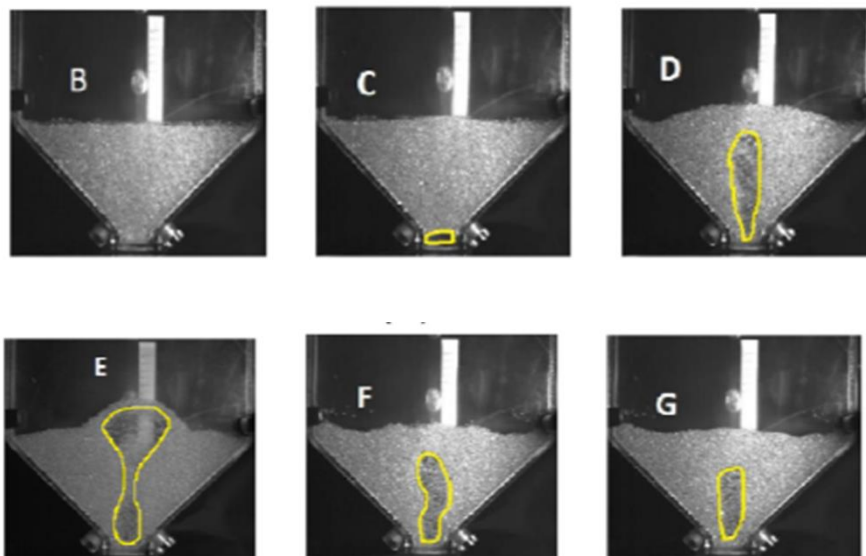
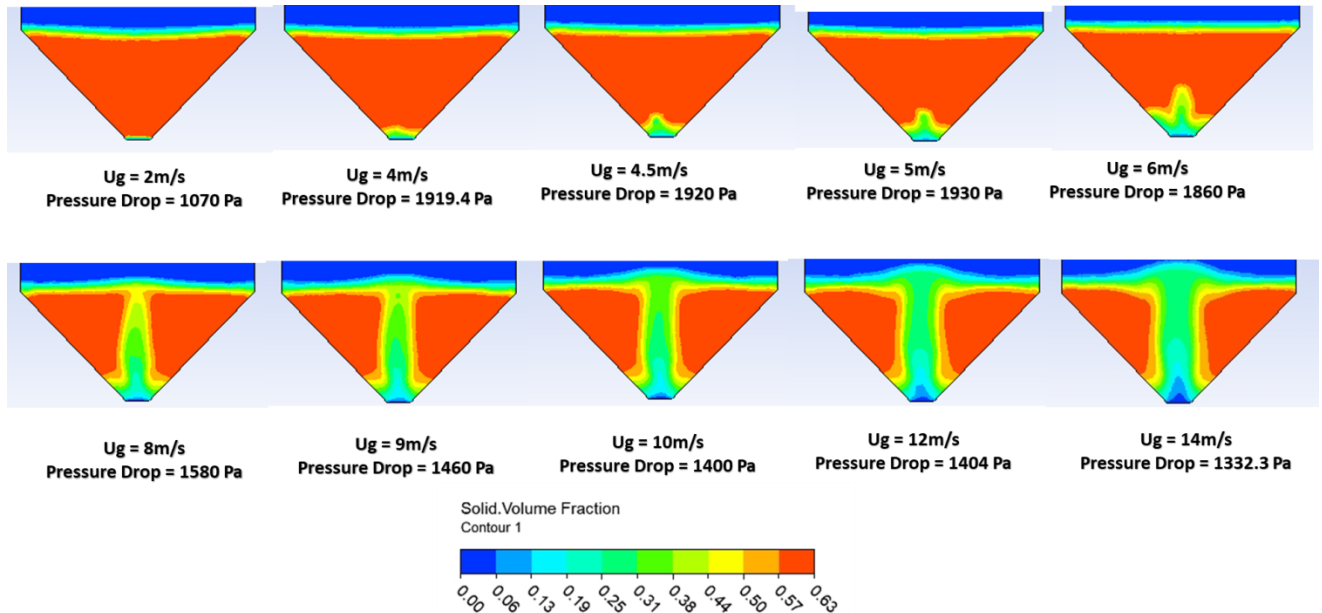


Figure 5.2: Experimental snapshot of the formation of cavity and particle distribution during the spouting process.



## 5.2 Spouting Phenomena



**Figure 5.3: Simulation results of mean volumn fraction contour**

The simulation results show contour plots of solid volume fraction at various gas velocities. As the inlet velocity increases, a clear spout channel forms at the center, with reduced particle concentration and increased particle uplift.

- At  $U_g = 2$  m/s, the bed is still dense with little spout formation.
- At  $U_g = 5$ – $6$  m/s, partial spouting begins, but it is unstable.
- At  $U_g = 10$  m/s and above, the spout becomes well-developed and symmetrical.

### Spouting Coherence

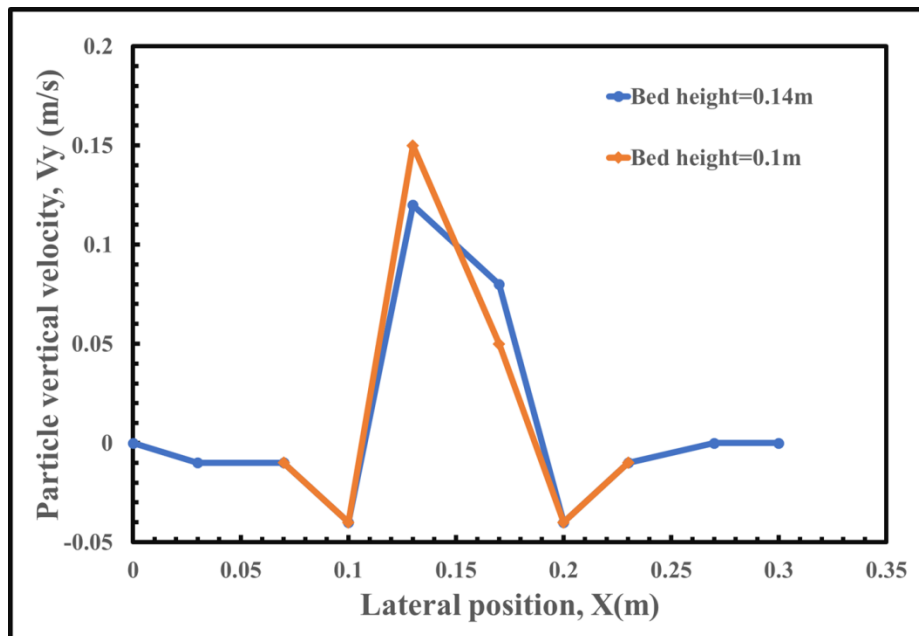
The observed results help differentiate between:

- Coherent spouting: symmetrical, steady spout and fountain

- Incoherent spouting: fluctuating spout edges, irregular particle ejection

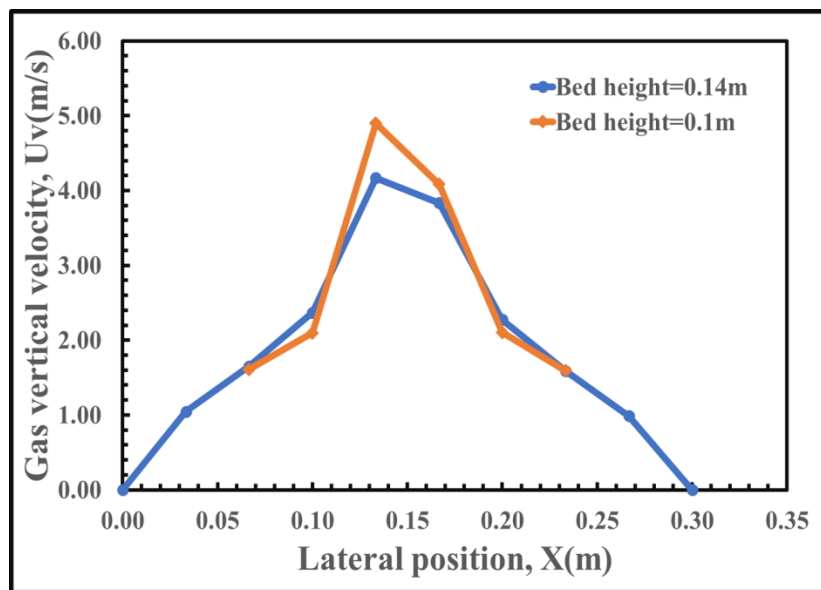
### 5.3 Spout Bed Dynamics

Detailed flow field analysis was performed through, Velocity profiles of gas and solid phases and Solid volume fraction distributions. Data extracted at two different vertical heights in the spouted bed  $Y=10\text{cm}$  and  $Y=14\text{cm}$ .



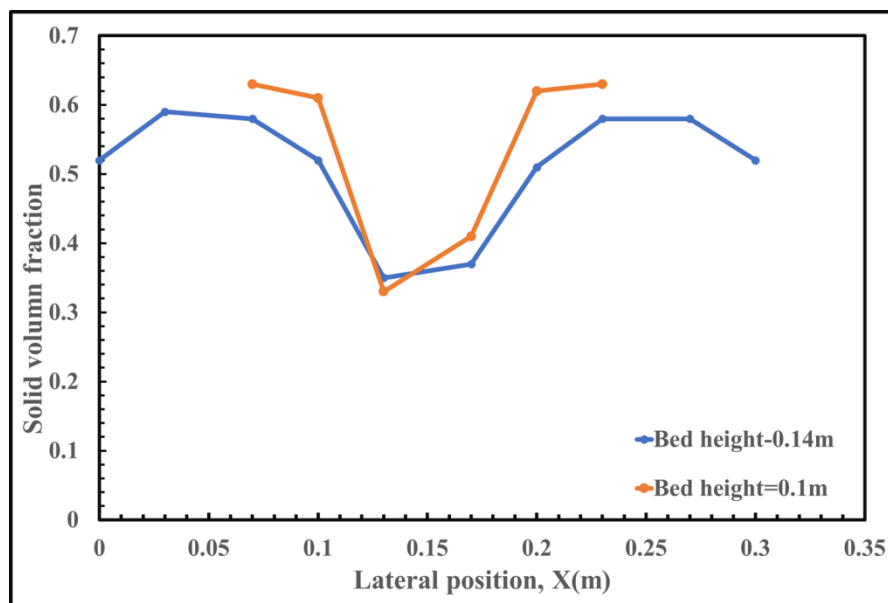
**Figure 5.4: Lateral distribution of solid vertical velocity**

Vertical velocity of solids shows a peak in the spout region and a gradual decrease toward the annular zone.



**Figure 5.5: Lateral distribution of gas vertical velocity**

Gas vertical velocity also shows a peak in the spout region and a gradual decrease toward the annular zone but with higher magnitudes due to lower density.



**Figure 5.6: Lateral distribution of solid volume fraction**

Solid volume fraction profiles show, Low values at the center indicating spout and high values at the annulus, where particles descend.

## 6 Comparison between CFD-DEM and TFM

In the numerical modeling of gas–solid multiphase systems such as spouted beds, two predominant computational approaches are widely employed: the Eulerian–Eulerian Two-Fluid Model (TFM) and the Eulerian–Lagrangian Discrete Element Method (CFD-DEM). Both methods aim to simulate the dynamics of gas–solid interactions, but they differ fundamentally in how they represent the solid phase, handle interphase interactions, and resolve particle-scale physics. The selection between these methods often depends on the required level of physical detail, computational cost, and the scale of the reactor being modeled [48].

The Two-Fluid Model (TFM) is a continuum-based approach where both the gas and solid phases are treated as interpenetrating continuous media. Each phase occupies a fraction of the volume, defined by the volume fraction  $\alpha_q$  and has its own set of governing equations for mass and momentum conservation. The interactions between phases are modeled through interphase momentum exchange terms—most commonly, the drag force, which is computed using empirical correlations such as the Gidaspow, Wen–Yu, or Syamlal–O'Brien models. TFM is best suited for systems where the number of particles is extremely large and the detailed resolution of individual particle trajectories is not feasible. In this approach, the influence of particle–particle collisions and granular behavior is incorporated indirectly through models such as the Kinetic Theory of Granular Flow (KTGF), which introduces solid-phase pressure, viscosity, and granular temperature based on the statistical behavior of particle assemblies.

On the other hand, the CFD–DEM (Computational Fluid Dynamics coupled with the Discrete Element Method) approach offers a more physically detailed

framework by modeling the gas phase as a continuum (Eulerian), while explicitly tracking each solid particle (Lagrangian) in space and time. Every individual particle is subjected to Newton's second law, considering contact forces (using models like the Hertz–Mindlin or spring-dashpot) and fluid-induced forces such as drag, lift, and buoyancy. The interaction between gas and particles is captured through local coupling based on the instantaneous particle positions and relative velocities. CFD–DEM thus allows for precise resolution of phenomena such as particle clustering, segregation, recirculation, and spout formation, which are difficult to capture in volume-averaged continuum models. However, this added accuracy comes at a significant computational cost, which scales with the number of particles—making CFD–DEM less suitable for industrial-scale simulations unless advanced hardware and parallelization techniques are used.

In the context of spouted beds—characterized by cyclic flow patterns, dense particle zones, and localized jets—the choice between CFD–DEM and TFM becomes particularly important. While TFM offers robust prediction of bulk flow behavior, mean velocities, and regime transitions with moderate computational demand, it may underpredict microscale dynamics such as coherent spout collapse or particle mixing at the spout-annulus boundary. In contrast, CFD–DEM can resolve these phenomena accurately by resolving individual particle motions and collisions but is often limited to small-scale or 2D configurations due to computational resource constraints. Therefore, the selection of a modeling approach must be made based on a clear understanding of simulation goals, desired accuracy, physical resolution, and available computing power [48].

## 6.1 Parameter Comparison

To compare the numerical modeling capabilities of the **Euler–Lagrangian (CFD–DEM)** and **Euler–Euler (TFM)** approaches in spouted bed simulations, a set of important physical, numerical, and performance-related parameters are presented below. These reflect differences in how each method treats the gas–solid flow system, as well as their computational behavior and output resolution.

| Parameters                                        | CFD DEM              | TFM                  |
|---------------------------------------------------|----------------------|----------------------|
| Bed dimension (pseudo 2D RSB)                     | 0.3m x 0.02m x 0.74m | 0.3m x 0.02m x 0.74m |
| static bed height, (Hs, m)                        | 0.14                 | 0.14                 |
| Minimum spouting velocity (Ums, m/s)              | 9.74                 | 10                   |
| Inlet superficial gas velocity (Ug/Ums)           | 1.2                  | 1.2                  |
| Particle density ( $\rho_p$ , kg/m <sup>3</sup> ) | 2526                 | 2526                 |
| Particle diameter ( $d_p$ , mm)                   | 2.58                 | 2.58                 |
| Particle friction coefficient                     | 0.3, 0.55 & 0.67     | ----                 |
| Particle restitution coefficient                  | 0.97                 | 0.97                 |
| Particle poisson's ratio                          | 0.35                 | ----                 |
| Particle Young's modulus (Pa)                     | 1x10 <sup>9</sup>    | ----                 |

|                   |                      |                    |
|-------------------|----------------------|--------------------|
| CFD timesteps (s) | $2.5 \times 10^{-5}$ | $1 \times 10^{-4}$ |
| DEM timesteps (s) | $2.5 \times 10^{-6}$ | ----               |
| Total timesteps   | ----                 | 30000              |
| Residual criteria | ----                 | $10^{-5}$          |

**Table 6.1 Parameter values comparison between CFD DEM and TFM**

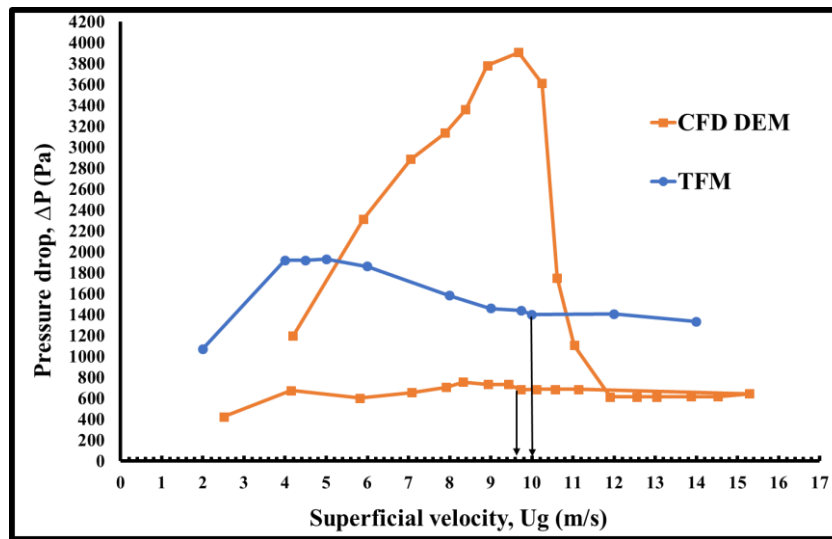
Importantly, the computational time and scalability differ significantly. CFD–DEM simulations are computationally expensive, scaling with the number of particles (often in the millions), and therefore are limited to short durations or small domains unless high-performance computing resources are available. On the other hand, TFM simulations can be run over larger geometries and longer time scales, making them more feasible for reactor-scale modeling and optimization tasks.

## **6.2 Graphical Comparison of CFD–DEM and TFM**

### **6.2.1 Pressure drop characteristics curve**

This graph presents the variation of pressure drop ( $\Delta P$ ) across the spouted bed with increasing gas inlet velocity ( $U_g$ ). Both CFD–DEM and TFM results are plotted for validation.





**Figure 6.1 Pressure Drop Characteristics Curve**

Observation:

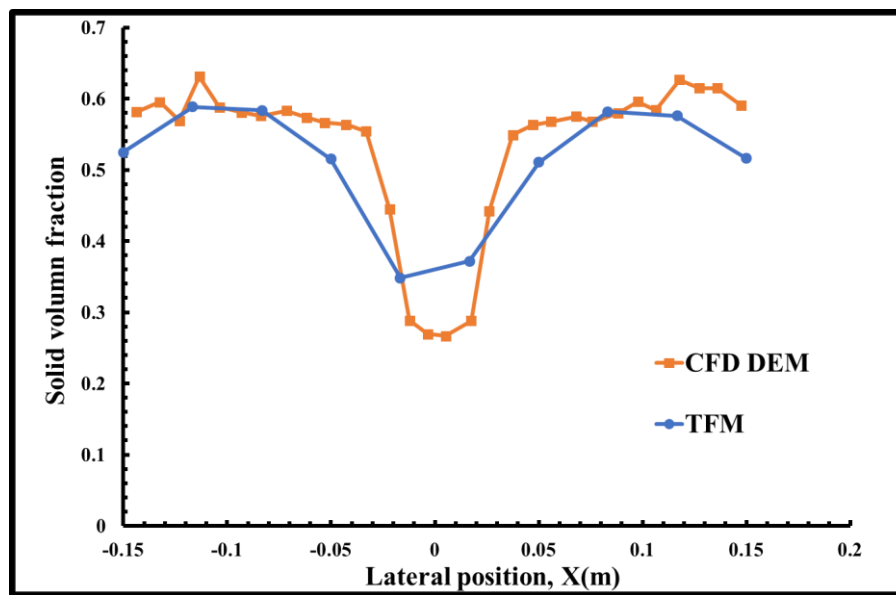
- CFD–DEM captures subtle fluctuations in pressure drop as the bed transitions from packed to spouting regimes.
- TFM also shows a similar trend but with a smoother curve due to volume-averaging of gas–solid interactions.

Explanation:

In CFD–DEM, local pressure variations from particle-level dynamics (e.g., collisions and voids) are preserved. In contrast, TFM computes pressure using a continuum approach, which smooths out microscale fluctuations.

### 6.2.2 Solid Volume Fraction vs Lateral Distance

The graph compares lateral distribution of the solid phase at a fixed bed height (14 cm) for both methods.



**Figure 6.2: Lateral distribution of solid volume fraction at bed height  $Y=1.0H_s$ , Superficial gas velocity  $U_g = 12$  m/s.**

Observation:

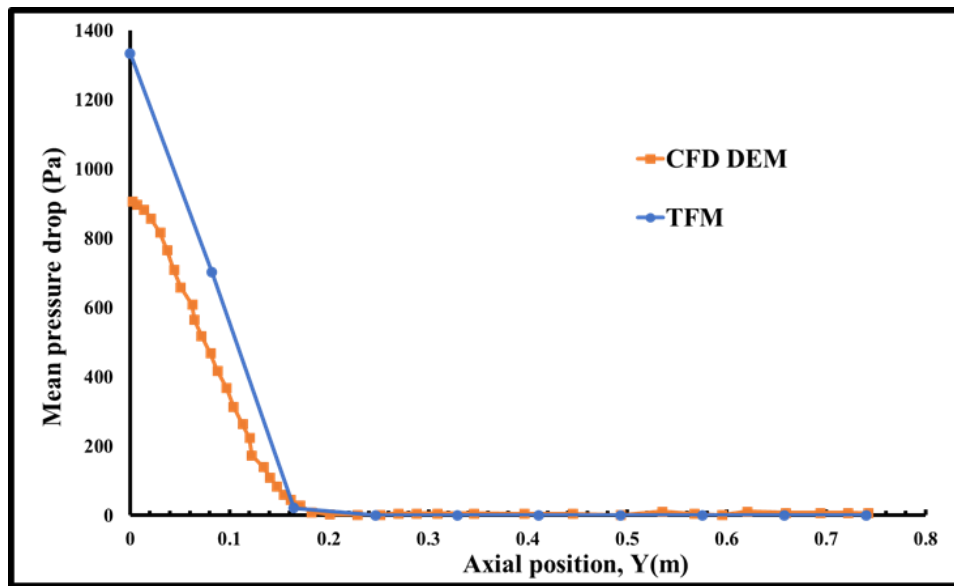
- CFD–DEM shows peaks and troughs indicating solid clustering and channeling, especially near the spout centerline.
- TFM predicts a smoother, symmetric profile across the bed width.

Explanation:

CFD–DEM resolves individual particle motion, making it sensitive to clustering or uneven flow in the spout and annulus zones. TFM averages particle effects over a mesh volume, leading to a more homogeneous solid volume fraction profile

### 6.2.3 Mean Pressure vs Axial Distance

This figure shows how mean pressure varies along the vertical axis of the spouted bed.



**Figure 6.3: Axial distribution of mean pressure at Superficial gas velocity,  $U_g = 12$  m/s.**

Observation:

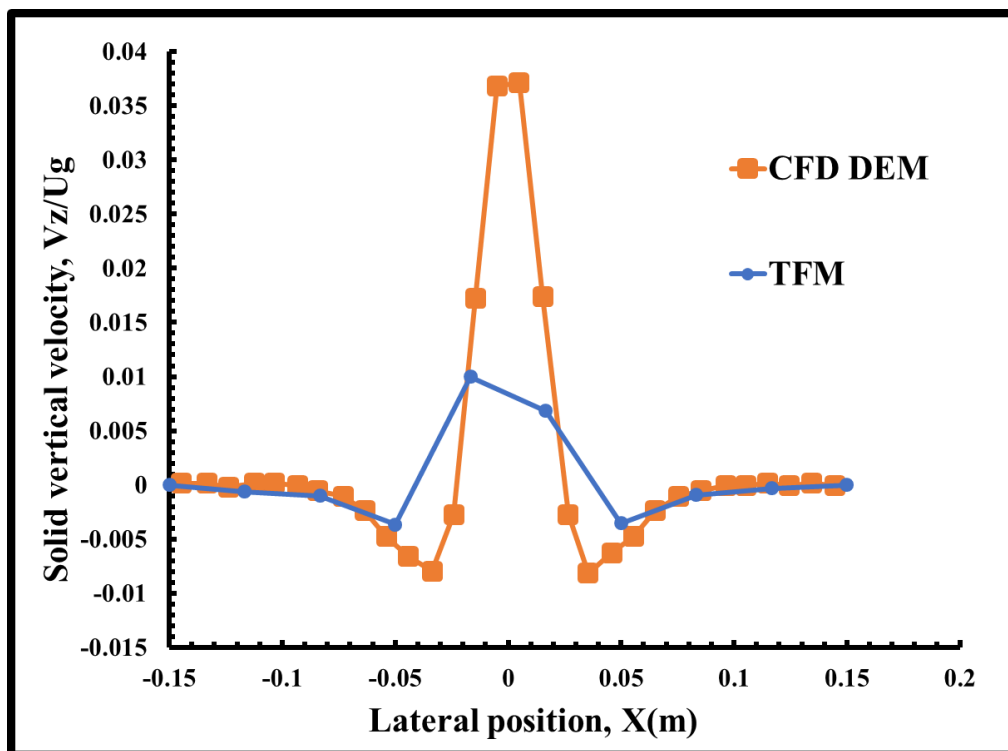
- Both models capture the axial gradient, but CFD–DEM reveals localized pressure dips due to instabilities or particle fallback.
- TFM shows a more linear, smooth pressure decline with height.

Explanation:

DEM captures transient events like particle jets collapsing or diverging. In contrast, TFM lacks this resolution due to its continuum assumption, but still reflects the mean trend accurately.

#### **6.2.4 Solid Vertical Velocity ( $V_z/U_g$ ) vs Lateral Distance**

The normalized vertical velocity of solids at 14 cm height is plotted against lateral bed position.



**Figure 6.4: Lateral distribution of solid vertical velocity at bed height  $Y=1.0H_s$ , Superficial gas velocity  $U_g = 12$  m/s.**

Observation:

- In CFD–DEM, a sharp velocity peak is seen near the centerline (spout core) with rapid drop-off to either side.
- TFM shows a wider distribution, with reduced peak value and more gradual decline.

Explanation:

The narrower peak in CFD–DEM implies better resolution of the core spout jet, while TFM captures only the averaged rise due to its mesh coarseness and KTGF-based modeling.

## **6.3 Reasons for Differences Between CFD–DEM and TFM**

The differences observed in simulation outputs such as pressure drop behavior, velocity distribution, spout shape, and solid volume fraction between the CFD–DEM and TFM models arise from several fundamental modeling assumptions and numerical implementations. These distinctions influence the resolution, accuracy, and physical realism of each method when applied to gas–solid spouted bed systems.

Below are the key reasons for the discrepancies between the two models:

### **1. Modeling Approach**

CFD–DEM:

Treats each particle individually (Lagrangian for solids, Eulerian for gas)

TFM (Euler–Euler):

Treats both gas and solids as interpenetrating continua and solves volume-averaged equations

### **2. Particle Interaction**

CFD–DEM:

Directly resolves particle–particle collisions using contact models (e.g., Hertz–Mindlin)

TFM:

Uses statistical models (e.g., kinetic theory of granular flow - KTGF) to approximate particle interactions

### **3. Resolution and Accuracy**

CFD–DEM:

Very high resolution — tracks real particle trajectories and captures micro-scale dynamics accurately

TFM:

Smoothed-out behavior — averages over many particles and cannot capture particle-scale fluctuations or clustering as well.

## **6.4 Discussion**

- In the present investigation,  $U_{ms}$  was found to be 9.74 from CFD DEM and 10 m/s from TFM.
- Trends for the average bed pressure drops for both the CFD DEM and TFM one are found identical; however, a quantitative difference might have originated from the initial packing configuration and due to the assumptions in ANSYS Fluent's Two-Fluid Model.
- Nevertheless, the TFM obtained spouting pattern is similar to CFD DEM results and  $U_{ms}$  also compares well with the CFD DEM results.

## 7 Validation with Ergun equation using Two-Fluid model

Packed beds are essential components in many industrial processes, such as fixed-bed reactors, filtration systems, and heat exchangers. They consist of stationary solid particles, through which a fluid flows, allowing for enhanced contact between phases and efficient transfer processes. In such systems, accurate estimation of pressure drop is critical for design, operation, and scale-up [49].

To quantify the pressure loss across a packed bed, one of the most widely adopted empirical correlations is the Ergun equation, originally proposed by Ergun in 1952. This equation expresses the pressure drop as a function of bed porosity, particle size, and fluid properties. It is applicable over a wide range of flow regimes by incorporating both viscous and inertial contributions to resistance:

$$\frac{\Delta P}{L} = \frac{150\mu U(1 - \varepsilon)^2}{\varepsilon^3 d_p^2} + \frac{1.75\rho U^2(1 - \varepsilon)}{\varepsilon^3 d_p}$$

Where:

- $\frac{\Delta P}{L}$  : pressure drop per unit length (Pa/m)
- $\mu$  : dynamic viscosity of fluid (Pa·s)
- $U$  : superficial velocity (m/s)
- $\rho$  : fluid density (kg/m<sup>3</sup>)
- $\varepsilon$  : void fraction or porosity
- $d_p$  : particle diameter (m)

This relation has become a standard tool for packed bed analysis due to its simplicity and reasonable accuracy for beds filled with spherical, uniformly sized, and non-porous particles at moderate Reynolds numbers [50] .

However, limitations exist, particularly when applying the Ergun equation to packed beds with complex geometries or porous particles, or when transitioning to numerical modeling environments like CFD using the Two-Fluid Model (TFM). The TFM treats the gas and solid phases as interpenetrating continua, solving separate sets of mass and momentum equations for each phase, and incorporating drag correlations (e.g., Gidaspow, Wen-Yu) to account for momentum exchange .

The application of TFM in ANSYS Fluent allows for detailed spatial and temporal resolution of hydrodynamics and pressure profiles in multiphase systems. However, since the TFM inherently assumes local averaging and ignores discrete particle interactions, validation of TFM predictions using classical correlations like the Ergun equation becomes necessary to confirm its fidelity under dense, static, or near-static conditions such as in packed beds [41].

In this chapter, we attempt a validation study by simulating a stationary packed bed configuration using ANSYS Fluent's Eulerian multiphase model and comparing the predicted pressure drop against values calculated using the Ergun equation. This involves matching boundary conditions, material and flow rates in the simulation and theoretical model. The consistency between simulated and theoretical curves is examined and reasons for any deviation are discussed [51].

## **7.1 Computational Model**

In this study, the computational modeling of gas–solid flow within a packed bed configuration was performed using the Eulerian–Eulerian Two-Fluid Model



(TFM) framework available in ANSYS Fluent. The TFM approach treats the gas and solid phases as interpenetrating continua and solves individual sets of mass and momentum equations for each phase. This allows the model to simulate phase interaction through momentum exchange terms, often governed by empirically derived drag laws.

To simulate a true packed bed condition in the Eulerian–Eulerian Two-Fluid Model (TFM), it is necessary to restrict the motion of the solid phase completely. In ANSYS Fluent, this was achieved by fixing the cell zone condition of the solid phase. Specifically, the X, Y, and Z velocity components of the solid phase were set to zero throughout the bed domain. This approach effectively "locks" the solid phase in place, preventing any translational motion and emulating a packed structure in which gas flows through stationary particles. This method is consistent with best practices in multiphase CFD for replicating static beds, where solids are not fluidized but resist flow, and all particle interactions are modeled via the drag force only [53]. This boundary control ensures that the solid volume fraction remains constant, and the system operates in a regime where pressure drop predictions from the Ergun equation are valid. It also ensures that there are no granular temperature effects, and the drag model dominates the interphase momentum exchange [52].

### **7.1.1 Drag Model: Gidaspow Model**

In this simulation, the Gidaspow drag model was employed to model the interphase momentum exchange between the gas and solid phases. The Gidaspow model is a hybrid formulation that switches between the Wen–Yu model for dilute regions and the Ergun equation for dense packed beds. This makes it especially suited for simulations that encompass both fluidized and packed regimes, as in our case [52].

The interphase momentum transfer coefficient expressed as:

$$K_{sq} = \begin{cases} \frac{3 c_D \alpha_q \alpha_s \rho_q |v_q - v_s|}{4 d_p} & \alpha < 0.2 \\ 150 \frac{\mu_g \alpha_s^2}{\alpha_q^2 d_p^2} + 1.75 \frac{\rho_q \alpha_s}{\alpha_q d_p} |v_q - v_s| & \alpha \geq 0.2 \end{cases}$$

Here the variables are defined as  $\alpha$  ,  $\rho$  ,  $v$  ,  $K_{sq}$  ,  $\mu$  and  $d_p$  volumn fraction of the particular phase, density, velocity, interphase momentum exchange coefficient, dynamic viscosity, diameter of particle and two phases Gas (q) and solid (s).

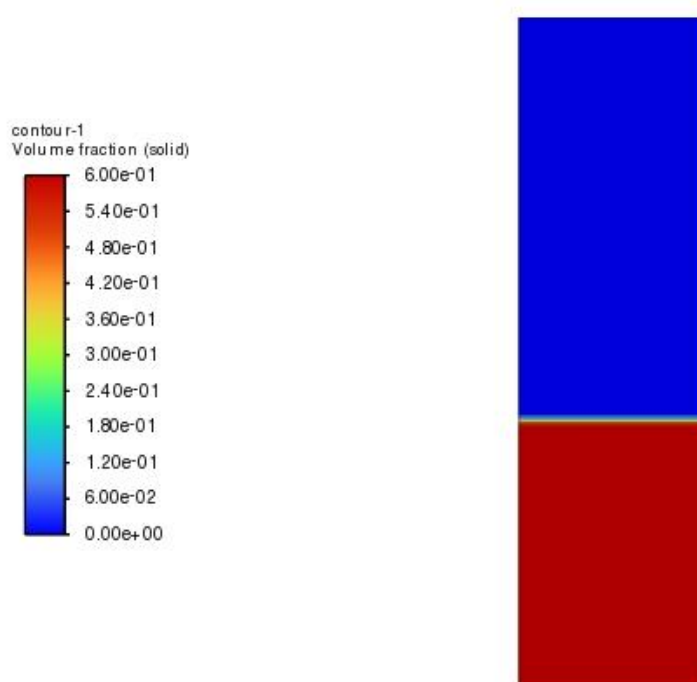
### 7.1.2 Simulation parameters

| Parameter                                         | Value              |
|---------------------------------------------------|--------------------|
| Particle density, $\rho_p$ (kg/m <sup>3</sup> )   | 2500 (glass beads) |
| Gas density, $\rho_g$ (kg/m <sup>3</sup> )        | 1.225              |
| Mean particle diameter, $d_p$ ( $\mu m$ )         | 275                |
| Restitution coefficient                           | 0.97               |
| Initial solid packing                             | 0.6                |
| Superficial gas velocity, $U_g$ ( $\frac{m}{s}$ ) | 0.01 – 0.065       |
| Bed width (m)                                     | 0.28               |
| Bed height (m)                                    | 1                  |
| Bed thickness (m)                                 | 0.025              |

|                          |           |
|--------------------------|-----------|
| Static bed height (m)    | 0.4       |
| Time step (s)            | 0.001     |
| Maximum no of iterations | 20        |
| Convergence criteria     | $10^{-3}$ |

**Table 7.1: Simulation model parameters**

These parameters are in agreement with prior validated CFD studies on gas–solid hydrodynamics in packed



**Figure 7.1: Geometry of packed bed column used in simulation**

## 7.2 Results

To simulate a packed bed in ANSYS Fluent using the Eulerian–Eulerian Two-Fluid Model (TFM), the motion of the solid phase was completely restricted by

applying fixed cell zone conditions. Specifically, the X, Y, and Z velocity components of the solid phase were set to zero in the solver's cell zone conditions panel. This configuration replicates a physically realistic static bed, where the solid phase remains immobile, and only the gas phase flows through the void spaces of the packed structure [54].

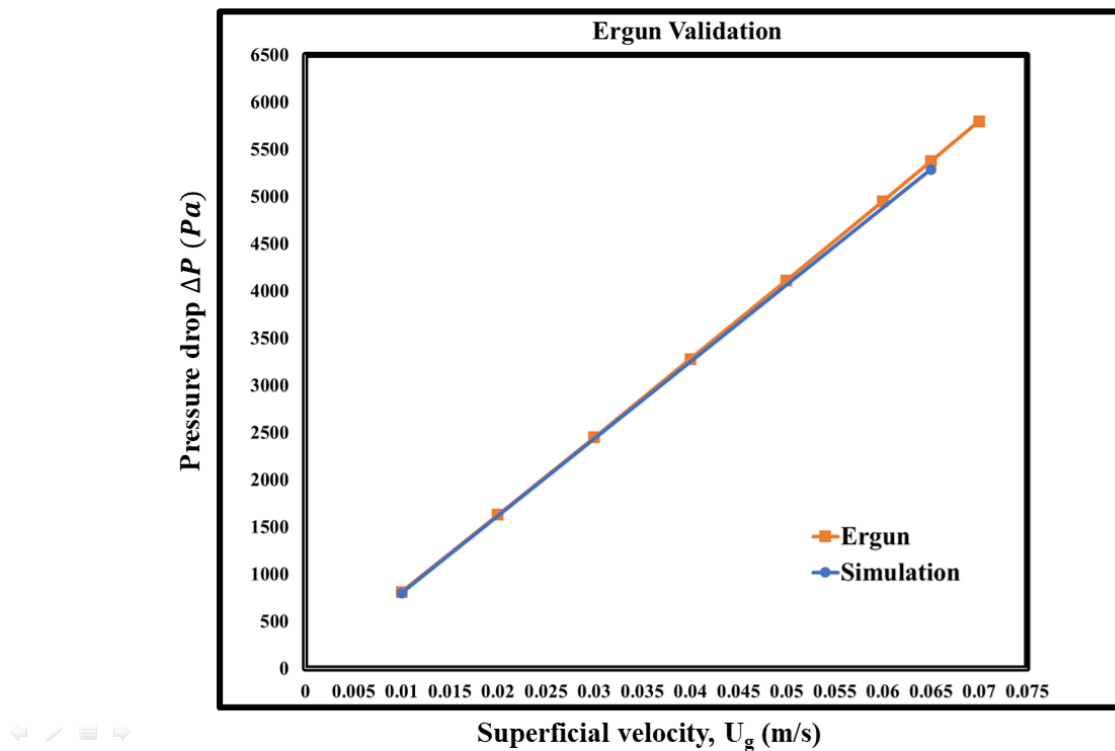
This method ensures that:

- The solid volume fraction remains constant throughout the bed
- Granular temperature modeling is unnecessary, since solids do not move
- Momentum exchange occurs only through drag, governed by the Gidaspow model, which incorporates the Ergun equation in dense regions

With this setup, the simulation domain effectively emulates a packed bed configuration, and the resulting pressure drop can be directly compared with theoretical predictions from the Ergun equation, which assumes a stationary solid matrix [55].

#### Validation of Pressure Drop vs. Superficial Velocity

To validate the CFD predictions, a range of gas superficial velocities ( $U_g$ ) was applied at the inlet, and the corresponding pressure drop ( $\Delta P$ ) was recorded across the bed for each case. These simulated results were plotted against theoretical pressure drops computed using the Ergun equation for the same porosity and particle diameter.



**Figure 7.2: Comparison of Pressure Drop Predicted by TFM vs. Ergun Equation**

The pressure drop vs. superficial velocity curve presented in Figure 7.2 shows a strong agreement between the TFM simulation results and theoretical values from the Ergun equation.

The maximum deviation was less than 5%, which validates that:

- Gidaspow drag model used in TFM accurately reflects pressure resistance in static beds
- Fixed solid zone condition approach is valid for simulating packed beds in Eulerian–Eulerian frameworks.

This confirms that the TFM framework in Fluent, when configured correctly, is capable of reproducing real-world packed bed behavior with high fidelity.

## 8 Meandering

In the realm of granular flow and fluidized systems, meandering instability refers to the undulating, wave-like paths taken by gas flow as it traverses through a packed or semi-fluidized bed of particles. It is a fascinating hydrodynamic behavior arising from the complex interactions between the injected fluid and the surrounding particulate matter. Such behavior is commonly observed not only in natural systems such as river channels but also in engineered environments like spouted beds, where local jets of gas are introduced into a granular medium [56].

The meandering of gas jets in a granular bed represents a transitional regime between straight laminar flow and chaotic or turbulent regimes. This phenomenon is of both fundamental and practical interest, as it exhibits a combination of fluid-like and solid-like behavior, termed fluid-solid duality [56]. In this dual regime, part of the granular bed behaves like a deformable solid, resisting the flow, while other regions become fluidized, allowing the gas to navigate through with reduced resistance.

In CFD simulations using Eulerian–Eulerian TFM (Two Fluid Model) approach, the meandering regime can also be captured, particularly when analyzing the evolution of solid volume fraction contours over time. The dynamic shift of high and low-density regions, visualized through solid volume fraction plots, provides an excellent representation of the meandering air paths in the medium. These simulations help validate experimental findings and allow deeper insight into the role of particle-fluid coupling, phase interaction, and instability development.

## 8.1 Contours of Meandering in Two-Fluid model

In this section, the dynamic evolution of meandering gas paths through a granular bed is investigated using the Eulerian–Eulerian Two-Fluid Model (TFM) in ANSYS Fluent. This modeling approach allows the gas and solid phases to be treated as interpenetrating continua, governed by conservation equations for mass and momentum. The meandering phenomenon, characterized by periodic lateral deflection of gas jets in granular beds, has been studied widely in experimental physics, but its replication through CFD remains an area of active research. In this study, EPS (Expanded Polystyrene) particles were simulated in a quasi-2D bed using transient multiphase flow modeling to capture time-varying flow structures.

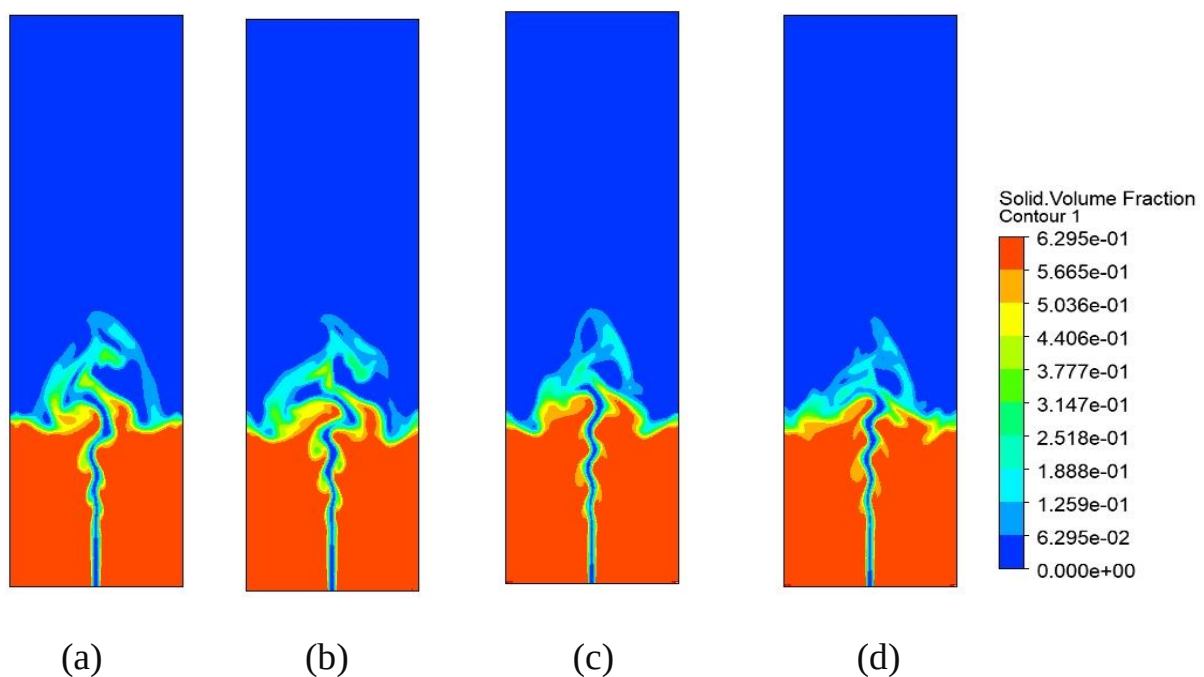
The simulation domain had the following dimensions:

- Column Height: 1 m
- Column Width: 0.3 m
- Thickness: 0.01 m
- Particle Density: 20 kg/m<sup>3</sup>
- Particle Diameter: 387.854 μm
- Inlet Diameter: 0.01 m
- Inlet Area:  $7.85 \times 10^{-5}$  m<sup>2</sup>
- Static Bed Height: 0.3 m
- Flow Rates Used:   1. 10 LPM (velocity: 2.1235 m/s)  
                              2. 15 LPM (velocity: 3.184 m/s)
- Flow Time Simulated: 10 seconds

The simulation used Gidaspow drag model, standard  $k-\epsilon$  turbulence model, and second-order implicit time formulation with a time step of 0.0001 seconds. The computational model was run using phase-coupled SIMPLE scheme, velocity-inlet boundary condition at the gas nozzle, and pressure-outlet at the top. The initialization was done using standard method with residual criteria of  $1e-5$ .

### Analysis of Meandering Contours

To visualize and analyze the development of meandering, contours of solid volume fraction were saved at regular time intervals. A total of four frames were captured at different simulation.



**Figure 8.1: Contour of solid volumn fraction (a) at 1second and 15lpm (b) at 1.1 second and 15lpm (c) at 1.3 second and 10lpm (d) at 1.4 second and 10lpm**

### Contour Comparison: Flow Rate = 15 LPM (3.184 m/s)

At 1 sec: A nearly vertical gas path begins to form through the static bed. Minor lateral deflection appears, indicating the onset of instability.



At 1.1 sec: The jet visibly deflects to one side, showing the early phase of sinusoidal meandering. The amplitude is small but perceptible.

At 1.3 sec: A mild meandering path is observed. However, the jet curvature is subtle compared to the higher flow case.

At 1.4 sec: The path remains confined with low amplitude, suggesting weaker instability and reduced meandering due to insufficient velocity.

## **8.2 Conclusion of Contour Analysis**

The simulation successfully replicated scale-invariant meandering patterns for EPS particles in a thin-bed configuration. The higher flow rate (15 LPM) clearly demonstrates stable sinusoidal paths, while lower flow rate (10 LPM) shows marginal jet penetration and negligible wave formation. These results not only validate the TFM's ability to simulate jet instabilities but also reflect strong agreement with experimental contours observed in literature .

The ability to capture meandering patterns using the Two-Fluid Model not only validates the predictive power of CFD tools but also opens avenues for future applications. Understanding meandering behavior is particularly valuable in optimizing reactor design, gas-solid flow control, and bed stability in industries such as fluidized bed combustion, drying, and coating. These findings can guide future experimental and simulation studies aimed at enhancing mixing uniformity and controlling flow regime transitions in particulate systems [56].

## 9 Conclusion and Future Work

### 9.1 Conclusions

This study successfully investigated the hydrodynamics of a 2D rectangular spouted bed using both experimental and computational approaches. The primary objective was to accurately determine the minimum spouting velocity ( $U_{ms}$ ) and understand the spout structure and flow dynamics under varying gas velocities.

A combination of experiments using glass beads and a custom-fabricated spouted bed, along with ANSYS Fluent simulations employing the Two-Fluid Model (TFM) and standard  $k-\varepsilon$  turbulence model, allowed for in-depth observation and validation of key flow parameters.

The following conclusions were drawn:

- The experimentally determined minimum spouting velocity was approximately 11.13 m/s, while the simulated value was 10 m/s, showing close agreement within an acceptable error margin.
- Pressure drop vs. gas velocity plots captured the fixed-to-spouting regime transition accurately and validated the simulation model.
- Contour comparisons of solid volume fraction from simulations and experimental video snapshots confirmed the formation of well-defined spout and annulus zones.
- Velocity profiles of gas and solid phases, and volume fraction distributions at different bed heights, matched classical spouted bed theory and showed consistent recirculation behavior.

- Grid independence was established with minimal deviation between 4 mm and 5 mm meshes, leading to the selection of the 5 mm grid for final runs to optimize computational time and resources.
- Coherent and incoherent spouting behavior was successfully identified and analyzed through both simulation contours and experimental observations. The transition from incoherent (unstable, asymmetric spout) to coherent spouting (symmetrical, stable flow) was captured clearly, validating the simulation model's ability to predict regime transitions.

The comparison between CFD–DEM and TFM clearly highlights the strengths and limitations of both modeling approaches. CFD–DEM offers high-resolution insights into particle-level dynamics, accurately capturing local instabilities, spout structure, and flow asymmetries. In contrast, TFM provides computational efficiency and smooth continuum predictions suitable for large-scale simulations, especially under dense flow conditions. While DEM better reflects transient granular phenomena, TFM remains a practical alternative when modeling time-averaged behavior. Overall, the choice between the two should be guided by the required accuracy, computational resources, and physical scale of the problem.

The validation study demonstrated that the Two-Fluid Model (TFM) in ANSYS Fluent, when configured with appropriate conditions and drag models, can accurately predict pressure drop across a packed bed. The simulated results closely matched the theoretical values from the Ergun equation, confirming the model's reliability under dense, stationary flow regimes. This establishes TFM as a robust tool for modeling packed beds and supports its use in more complex multiphase systems.

The simulation successfully captured the onset and evolution of meandering gas jets in a granular bed using the Two-Fluid Model. Distinct sinusoidal patterns

were observed at higher flow rates, consistent with experimental findings and theoretical predictions. The results confirm that TFM can effectively replicate meandering behavior and provide valuable insights into flow regime transitions and instability development in gas–solid systems.

## **9.2 Future work**

### **➤ CFD–DEM Modeling**

The Eulerian–Eulerian TFM approach assumes continuum treatment of the solid phase. Future studies can use CFD–DEM (Discrete Element Method) for particle-level tracking to capture collisions, segregation, and micro-scale dynamics with higher accuracy.

### **➤ Meandering**

Future studies can focus on exploring 3D meandering behavior, extending the work to include coherent–incoherent regime transitions and incorporating advanced turbulence or multiphase models could further improve the understanding and control of jet instabilities in industrial granular systems.

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